*The Clock Room and the Macro-Cognitive Accounting Framework: A Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms*, a five-volume book (in both Chinese and English editions), is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

License summary: https://creativecommons.org/licenses/by-nc-nd/4.0/deed.en Legal code: https://creativecommons.org/licenses/by-nc-nd/4.0/legalcode.en

Separately, pursuant to Article 4(3) of Directive (EU) 2019/790, the author expressly reserves the right to reproduce and extract this work for the purposes of text and data mining, including text and data mining for the training, fine-tuning, or evaluation of artificial intelligence or machine-learning models. This reservation is intended solely to opt out of the exception or limitation provided for in Article 4(1). It does not affect Article 3 or any other applicable exception or limitation, nor does it reduce the rights granted under the CC license.

Uses that fall outside the scope of the CC license and require permission under applicable law require the author's prior written permission, obtainable at theclockroom@139.com.

When sharing this work under the license, you must provide appropriate attribution and a link to the license, indicate whether modifications were made, and not suggest that the author endorses you or your use. The work may be used only for noncommercial purposes, and adapted material may not be shared.

All rights not expressly granted under the license are retained by the author. Third-party materials in this book, if any, are governed by their original rights statements or licenses and are not deemed to be additionally licensed by the author of this book.

Content Disclaimer

The content of this book is academic research and thought experiments; its analysis is based primarily on historical data and published literature. The author has twelve years of experience in macro strategy research at financial institutions and currently holds no securities investment advisory license. This book does not constitute investment advice, technical implementation advice, or policy advice, nor a recommendation or an offer with respect to any security, financial product, or trading strategy. Readers should exercise independent judgment and bear the risks of their own decisions. Past performance is not indicative of future results.

# Chapter 22: The Governance History of Financial Cognitive Debt—A Financial Historical Analysis under the MCAF Framework

## Abstract

Among the sample domains examined in this chapter, the financial system is one of the few that has, on a scale of centuries, evolved a relatively complete "cognitive debt processing mechanism," and it is currently a case suitable for long-horizon institutional tracking. From the establishment of the Philadelphia Contributionship in 1752 to the institutionalization of the Orderly Liquidation Authority under the Dodd-Frank Act in 2010, every major loss event selected in this chapter from 1752 through 2010 pushed forward the establishment or reinforcement of a node within the **compensation loop**. Using the MCAF framework (Macro-Cognitive Accounting Framework) as its analytical tool, this chapter asks the question of **dynamic debt tracking** in the institutional evolution of financial history: How does cognitive debt accumulate, at what speed does it manifest, and through what mechanisms is it liquidated? The division of labor between this chapter and Chapter 17 is as follows: Chapter 17 uses the Fiber Framework for **static structural diagnosis** (the Fiber Framework being the mathematical toolkit of Chapter 4—given a reading, the size of the set of true states measures the discrimination of the reading; mapping chain length, recursion ratio, and pseudo-fibers are its structural indicators), whereas this chapter uses the MCAF for **dynamic debt tracking** (manifestation speed, clock ratios, intervention effectiveness). The contrapuntal relation between this chapter and Chapter 15 is that finance is one of the few domains outside AI that already possesses a relatively complete compensation loop, thus providing an institutional baseline of

comparison for the diagnosis of the "missing manifestation chain" in §15 (AI lacking an institutionalized chain converting shadow debt into observable losses—disclosure, accountability, compensation, and related links). The incremental contribution of MCAF in this chapter does not lie in discovering historical facts unknown to the existing financial literature, but in **reorganizing existing findings through a unified six-variable + three-clock framework**, making possible structural comparisons across cases, such as Japan in the 1990s vs. the United States in 2008 vs. China in 2018–2026.

The methodological criterion of this chapter, carried over from Chapter 15, is: every structural diagnosis must pass the test—**after removing the MCAF variables, can the same judgment still be reached from this passage? Only if it cannot is the framework doing real work.** Four cases in financial history, Japan in the 1990s, the United States in 2008, China's New Asset Management Regulations in 2018, and China's quantitative-trading regulation (since 2015), are not laid out here according to a grid template of "complete six-variable mapping for each case" because that part has already been covered extensively in the existing literature. The cases appear in three places: for the time scales of the clock scenarios in §1, in the institutional history of the compensation loop in §2, and as the principal evidence for each of the three MCAF diagnoses in §3; quantitative-trading regulation serves only as the speed-scale illustration of §1.3 and is not developed separately. The increment of this chapter does not lie in "retelling financial history with MCAF," but in "extracting from financial history the structural diagnoses that MCAF can contribute." The three diagnoses are: the critical phase transition of the manifestation function $M(\tau)$, with the United States in 2008 as the primary evidence; the impossibility of pure defense-side intervention, with Japan in the 1990s as the primary evidence; and the feasibility of structural intervention and a progressive governance clock, with China's New Asset Management Regulations in 2018–2026 as the primary evidence. The governance implications of extreme clock ratios, Scenario III, namely the financial counterpart of precision speed-limiting, appear in §7 of Chapter 8.

## Keywords

Cognitive Reynolds Number; Cognitive Debt; Shadow Debt; Three Clocks; Compensation Loop; Manifestation Function; Structural Intervention; Speed-Limiting Intervention; Bimodal Distribution of Governance Clocks

---

## §0 The Logic of the Meta-Formula in Financial Scenarios

### §0.1 The Specific Mapping of the Six Variables in Financial Scenarios

The meta-formula established in Chapter 3 is the cognitive-debt identity $\Delta D_c = C \cdot \Lambda \cdot \phi - A \cdot E_c \cdot \|L\|$ (a difference form, a discipline of classification rather than a solvable equation); Chapter 3 also gives the individual simplified form of the Cognitive Reynolds Number, $Re_c \approx \Lambda / E_c$, which Chapter 7 unfolds into the six-variable full form:

$$Re_c^{AI} = \frac{C \cdot \Lambda \cdot \phi}{A \cdot E_c \cdot \|L\|}$$

where the three variables in the numerator measure the intensity of "projection," and the three variables in the denominator measure the intensity of "defense." This chapter takes the full form as its point of departure. The specific mapping of the six variables in financial scenarios is as follows.

**Unsealed inventory ($C$)** refers to the asset classes and risk-exposure types unsealed by financial innovation: subprime mortgages in the 1980s, over-the-counter derivatives in the 1990s, collateralized debt obligations (CDOs) and CDO-squared ($CDO^2$) in the 2000s, and algorithmic stablecoins and Direct Market Access (DMA) leverage channels in the 2010s. The key property of $C$ in finance is not scale but **nesting depth and irreversibility**: each wave of innovation is superimposed upon the uncleared exposures of the previous wave. Financial innovation does not automatically eliminate existing exposures; the stock can be reduced through normal repayment, contract maturity or termination, asset disposal, and restructuring, while the portion in which losses have already occurred must be recognized and assigned to a bearer. The financial scenario can declare observable proxy indicators for unsealed inventory ($C$)—total credit/gross domestic product (GDP) and derivatives notional principal/GDP—which, per the layering of Chapter 3 §7, are within-scenario relative-scale indicators, not physical units of measurement. This marks an important difference in observability from $C$ in the AI domain, namely unsealed generative capability, memorization capability, and agency.

**Media conductivity coefficient ($\Lambda$)** materializes in finance as leverage ratio, layers of securitization, derivatives notional multipliers, cross-border capital-flow velocity, and a trading-speed dimension captured by execution frequency (or the reciprocal of execution latency). The uniqueness of $\Lambda$ in financial scenarios lies in the fact that it can be artificially elevated by legal structures, Special Purpose Vehicles or SPVs, trusts, and asset-management nesting, without changing the true risk of the underlying assets. This is a structural feature that has no complete counterpart in the AI domain. The extreme values of $\Lambda$ reach physical limits in the context of high-frequency trading, at the millisecond-to-microsecond level, which is the most extreme projection speed that human governance can encounter in any scenario.

**Narrative load ($\phi$)** is the narrative component attached to price readings. Three core narratives that recur throughout financial history constitute the main content of $\phi$: **"this time is different"**, the self-justification of every bubble; **"too big to fail"**, the expectation of rescue for systemically important institutions; and **"a certain asset will never fall"**, as in Japan's belief in the 1980s that land prices would never fall, the United States' belief in the 2000s that housing prices nationwide would not fall simultaneously, and China's implicit-guarantee expectation in the 2010s. $\phi$ is not "a collection of false information"; each of these narratives has a partial empirical basis. What is truly dangerous is the structural operation of elevating **conditional historical regularities into unconditional axioms**. The high credit ratings before the crisis and the massive write-downs during it display the vast deviation between the risk assessment and the actual losses of some structured products—the gap between the 0.14% default probability implied by the AAA rating of senior CDO tranches and the massive write-downs that subprime CDOs actually suffered (researchers at the Federal Reserve Bank of Philadelphia estimate total write-downs at about 65% of the original issuance balance, Cordell et al., 2012) is one example. But per Chapter 3 §5.1, the value of the narrative leverage ratio ($\Omega = \phi/\|L\|$) may be referenced only after the scenario has completed scale definition and normalization—the rating-implied default probability and the ex post write-down ratio belong to two different calibers, and this chapter does not use their gap to estimate directly the order of magnitude of $\Omega$.

**Meta-audit capability ($A$)** is divided in finance into three levels: macro-audit ($A_{macro}$), cross-national stability assessments by the BIS and IMF; meso-audit ($A_{meso}$), the day-to-day supervision of domestic institutions by national regulatory authorities; and micro-audit ($A_{micro}$), institutions' internal risk control and investors' understanding of the products they hold. Finance has two structural advantages in $A$ relative to AI: first, the mandatory financial-statement disclosure system; second, the central bank's real-time data access to the banking

system. But these two advantages are neutralized in two circumstances: first, nesting at the $\Lambda$ layer renders the underlying assets invisible to external audit, as in the 2008 CDO case; second, the cognitive ceiling of regulators themselves is breached by product complexity, as in Greenspan's 2008 admission that his model of the world contained a flaw and that he still did not fully understand why the market mechanism had collapsed, cited in §3.1.5.

**Human biological substrate ($E_c$)** is a constant. This chapter treats the roughly 10 bits/s figure as an order-of-magnitude reference for the individual's **deep conscious-processing** information throughput (Zheng & Meister, 2025)—it is a lower-bound reference for institutional audit capability, not an upper-bound estimate: division of labor, professional training, and automation tools can still change meta-audit capability ($A$, Chapter 3 §3.1). The complexity of financial product documents and structures has grown far faster than individual cognitive processing capacity, and the ratio of the human biological substrate $E_c$ relative to the accumulation of $C \cdot \Lambda$ has therefore continuously deteriorated—a deterioration unrelated to increases in the number or training level of regulatory personnel.

**Legitimacy base ($\|L\|$)** operates according to the two-level aggregation of Chapter 3 §2.6: the hierarchical dimension (macro/meso/micro) and the typological dimension form a three-layer × three-type matrix—the typological dimension being performance legitimacy ($L_{perf}$), derived from sustained economic growth and asset-price appreciation; ideological legitimacy ($L_{ideo}$), deep beliefs such as "market efficiency" and "market self-regulation"; and procedural legitimacy ($L_{proc}$), derived from procedural devices such as the central bank as lender of last resort, deposit insurance, bankruptcy law, and orderly liquidation mechanisms. Within each layer the three types combine into a composite value by scenario-specific rules, and across layers the minimum is taken—overall bearing capacity is determined by the weakest layer; within a layer the three types can, in normal times, compensate for one another to a limited extent and are not mutually independent. Compared with the AI scenario, finance's $\|L\|$ has become relatively stable in the **procedural dimension** after 258 years of institutional evolution, which is exactly the core content of the compensation-loop narrative in §2. Shadow debt, $D_c^{shadow} = D_c^{true} - D_c^{obs}$, has **institutionalized manifestation channels** in finance, deposit-insurance triggers, rating downgrades, margin calls, and bankruptcy liquidation. This is the key structural difference in $\|L\|$ between finance and AI.

### §0.2 The Instantiation of the Three Clocks in Financial Scenarios

The three clocks of the MCAF framework exhibit a **distinctive contextual distribution** in financial scenarios. This is not an added complexity of the framework in the financial domain, but a structural feature formed by the institutional evolution of finance itself.

**Techno-capital clock ($\tau_c$)** is not a single value in financial scenarios, but presents three orders of magnitude according to the financial subdomain. The $\tau_c$ of credit-expansion financial activities is measured in years: during Japan's bubble period, the share of real-estate loans in total lending by major-city banks rose from about 5.5% in 1982 to over 12% in 1989 (Dinç & McGuire, 2004). The $\tau_c$ of derivative-creation financial activities is measured in months: CDO issuance grew from about $157 billion in 2004 to about $552 billion in 2006 (Benmelech & Dlugosz, 2009, reproducing the original statistics of the Securities Industry and Financial Markets Association, SIFMA), more than a threefold increase in two years; Credit Default Swap (CDS) notional amounts grew from about $8.4 trillion at the end of 2004 to about $62 trillion at the end of 2007 (ISDA, 2010), a more-than-sevenfold increase in three years, with an average doubling time of about one year. The $\tau_c^{exec}$ of high-frequency-trading

financial activities is at the millisecond-to-microsecond scale. This is the most extreme projection speed that human governance can encounter in any scenario, and it is also the core data point of §1.3 in this chapter. The analysis of precision speed-limiting paths under extreme clock ratios appears in Chapter 8, §7.

**Governance clock ($\tau_g$)** exhibits a **bimodal distribution** in financial scenarios, a structural feature of financial history that the MCAF framework can identify.

- **First peak: normal legislative mode.** The U.S. Dodd-Frank Act took about two years from the 2008 crisis to its signing in July 2010, about three years from signing to the final release of major rules such as the Volcker Rule in 2013, and about five years to the main compliance node in 2015; counting from the 2008 crisis, roughly five years to 2013, seven years to 2015, and eleven years to the arrival of Basel III arrangements in 2019—these nodes correspond to different rules and do not represent a single regulatory program running continuously for the same length of time. The EU's Markets in Financial Instruments Directive II (MiFID II) took about seven years from proposal in 2011 to full implementation in 2018. The $\tau_g$ of the normal legislative mode falls within the interval of 2–10 years.
- **Second peak: crisis-driven mode.** September 2008 is the canonical example. After Lehman filed for bankruptcy on September 15: the AIG bailout was announced on September 16 (the Federal Reserve authorized a secured loan of up to $85 billion, with total government support later expanding to about $182 billion); the Paulson draft of the Troubled Asset Relief Program, TARP, was submitted to Congress on September 19; the House rejected it on September 29; the Senate passed a revised version (74–25) on October 1; the House passed the Senate's revised version (263–171) on October 3, signed into law by Bush the same day; and the FDIC liquidity-guarantee program was launched on October 14. From the crisis point to legislative passage took about 18 days, from legislation to the landing of concrete tools took weeks, and the Federal Reserve's liquidity injection operated on the scale of hours. The $\tau_g$ of crisis mode falls within the range of days to weeks.

The bimodal distribution is not a matter of a "normal mode + exceptional deviation" of $\tau_g$; both peaks are structural parameters of the $\|L\|$ base. The compressive capacity of $\tau_g$ in crisis mode is not a product of deliberate design, but an **emergency response channel** preserved by the contract infrastructure accumulated over centuries at the meso layer, namely the lender-of-last-resort system, sovereign credit guarantees, and payment-clearing networks.

- **Third category: progressive governance (Chinese context).** The 2018 New Asset Management Regulations are the typical case: from the exposure draft in November 2017 to official release in April 2018 took about five months—this is the response cycle in the sense of the governance clock ($\tau_g$); the subsequent transition period (through the end of 2020, extended to the end of 2021 because of the pandemic) and the continued release of supporting rules (still being optimized through 2026) belong to the **governance persistence period** after the response—the duration of sustained implementation is not itself the response cycle, and this chapter distinguishes the two by descriptive naming. This is not a case of "poor execution." As §3.3 will argue, the governance persistence period is an institutional design made in stages and recalibrated under exogenous shocks, aimed at controlling the climbing slope of $M(\tau)$.

**Debt clock ($\tau_d$)** is the characteristic delay from the accumulation of cognitive debt to its first manifestation (Chapter 3 §3.2)—what returns to zero at triggering is the remaining waiting time, and repair after manifestation does not belong to the measure of $\tau_d$. For this reason this chapter separately uses "post-manifestation settlement

horizon" (a descriptive name, not a new symbol) to mark the long tail in which losses continue to be paid out in the economy after manifestation is complete: Japan, about 14–25 years, from the 1991 bubble burst to the late 2000s, with total factor productivity (TFP)-path deviation and labor-structure rigidification still being cashed out; the United States, about 10–15 years, from 2008 to the early 2020s, in the recovery of labor-force participation and long-term damage to affected communities; and the path of China's New Asset Management Regulations, expected to be 8–12 years and still ongoing. That the debt clock ($\tau_d$) and the governance clock ($\tau_g$) are **not synchronized**—rapid policy launch does not mean the repair of already deposited losses is completed in step—is the core diagnosis of §3.2 in this chapter: financial repair does not equal the end of crisis.

**The incremental positioning of MCAF in this chapter.** The incremental value of MCAF here does not lie in discovering historical facts or policy mechanisms unknown to the existing literature, but in **reorganizing existing findings through the unified six-variable + three-clock framework**, so as to make cross-case structural comparison possible; the point-by-point comparison of the three diagnoses with the existing literature (Gorton's and Brunnermeier's descriptions of crisis mechanisms, Caballero et al.'s zombie-lending model, and the policy articulations of progressive governance by Yi Gang, Guo Shuqing, and others) appears in the "Relation to the existing literature" sections at the end of each diagnosis in §3. The value of MCAF does not lie in discovering new cases, but in providing a **structural language that makes cases comparable across cases**. This language allows Japan in the 1990s, the United States in 2008, and China in 2018–2026 to be understood within the same coordinate system as three variants of governance-clock mode, rather than as three isolated policy narratives.

---

## §1 Financial Clocks vs. Governance Clocks: Four Scenarios

The clock ratio $\tau_g/\tau_c$ between the techno-capital clock $\tau_c$ and the governance clock $\tau_g$ exhibits four typical distributions in financial history. Each distribution corresponds to a dynamics of cognitive debt. This is the **first kind of structural observation** that MCAF can make about financial history; organizing cases by clock ratio is the specific perspective of this chapter. This section provides only clock observations and MCAF implications; it does not unpack the six variables. Cases appear here as ratio data points, while full diagnosis belongs in §3.

### §1.1 Scenario I: The Normal Accumulation Window $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$

**Data point**: the expansion period of U.S. shadow banking from 2000 to 2007.

- Techno-capital clock $\tau_c$ (doubling time of monthly CDO issuance): about 1–1.5 years.
- Techno-capital clock $\tau_c$ (doubling time of CDS notional amount): about 1 year.
- Governance clock $\tau_g$ (duration of regulatory response): from the SEC's approval in 2004 of the relaxation of net-capital rules allowing higher leverage for investment banks to the signing of the Dodd-Frank Act in 2010, about 6 years; to the release of the key rule (the Volcker Rule) in 2013, about 9 years; to the main compliance node in 2015, about 11 years.
- **Ratio** $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$ (regulation measured in years vs. market doubling measured in months to years).

**MCAF implication**: this is the **normal accumulation window** of shadow debt ($D_c^{shadow}$). Within the interval $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$, the projection side can complete 3–10 rounds of techno-capital-clock-scale innovation and expansion while the defense side still remains within the same governance-clock cycle. Each round of

expansion adds more unmanifested true cognitive debt, layer upon layer of CDO nesting, expansion of CDS notional scale, cross-border capital inflow, while observable debt $D_c^{obs}$ remains in the "compliant" state on regulatory balance sheets. The temporal mismatch between projection and defense is not merely a quantitative expression of "regulatory lag"; it is a **structural window** in which shadow debt is institutionally allowed to accumulate.

**What this section does not do**: it does not unpack the six-variable mapping of the 2000–2007 expansion period, unsealed inventory $C$, media conductivity coefficient $\Lambda$, narrative load $\phi$, meta-audit $A$, the human biological substrate $E_c$, and legitimacy base $\|L\|$. For the six-variable mapping see the survey by Sufi & Taylor (2022) and the existing analysis in §17; this section provides only the observation of clock ratio.

**Other historical ratio points**: for Japan's bubble period from 1985 to 1990, the available materials do not suffice to estimate the clock ratio under a unified caliber, and this chapter does not report a point estimate for it (the two ends of "bubble duration of 5 years vs. the interval until the first systemic response in the mid-1990s" do not define the starting points of market-expansion and governance-response cycles under the same caliber); in China's shadow-banking expansion in the 2010s, $\tau_g/\tau_c$ was about 0.9–1.6, shadow-banking doubling time about 2.5–3.5 years vs. response via the New Asset Management Regulations in 3–4 years. This is one of the historical cases among the four scenarios in which $\tau_g/\tau_c$ is closest to 1.

### §1.2 Scenario II: Crisis Compression Mode $\tau_g \approx 1$ day

**Data point**: the timeline of the U.S. crisis response in September 2008 was as follows:

- September 15: Lehman Brothers files for bankruptcy protection.
- September 16: the Federal Reserve announces the authorization of a secured loan of up to $85 billion to AIG through the New York Fed.
- September 19: Paulson submits the TARP draft to Congress, about 3 pages.
- September 29: the House rejects TARP for the first time, vote 205–228; the Dow Jones falls 777 points, about 7%, that day.
- October 1: the Senate passes the revised TARP legislation (74–25).
- October 3: the House passes the Senate's revised version (263–171), signed into law by Bush the same day.
- October 14: the FDIC liquidity-guarantee program is launched.
- **The governance clock $\tau_g$ from crisis point to legislative passage was about 18 days; the activation of Federal Reserve liquidity tools occurred on the scale of hours.**

**MCAF implication**: this is a mode of **emergency governance unique to finance**. The AI domain does not possess this compressive capacity, as §15.4 has already argued. In crisis mode, the governance clock $\tau_g$ is compressed from the normal scale of years to the scale of days, a **difference of roughly two orders of magnitude**. This is not because regulators suddenly become more diligent, but because the contract infrastructure accumulated at the meso layer, lender-of-last-resort institutions, sovereign credit guarantees, payment-clearing networks, and the preexisting congressional procedures on which TARP legislation relied, is structurally activated when crisis is triggered.

This compressive capacity has three structural preconditions. First, an **independent central-bank toolbox**: the Federal Reserve can inject liquidity directly without waiting for congressional legislation, a result accumulated

through the Federal Reserve Act of 1913 and the following century. Second, an **accelerated legislative mechanism**: TARP took only two days from first rejection in the House to passage of the revised version in the Senate, and four days in total to final enactment on October 3, which depended on preexisting emergency legislative procedures in Congress. Third, **protection by ideological legitimacy**: although the "too big to fail" narrative is criticized in normal times, during crisis it provides instant legitimacy at the level of ideological legitimacy $L_{ideo}$ for rescue. **None of these three preconditions currently exists in AI governance**: there is no "AI central bank" toolbox independent of legislative procedure, no accelerated legislative mechanism for AI crisis, and no ideological consensus that AI institutions are "too big to fail."

**Crisis compression does not change the post-manifestation settlement horizon.** This is a key structural observation that MCAF can identify: compressing $\tau_g$ from years to days does not shorten the long tail in which losses continue to be paid out in the economy after manifestation is complete. TARP and the AIG rescue in 2008 prevented total financial collapse within weeks, but U.S. labor-force participation continued to decline after 2008 and did not stabilize until 2015 (Hipple, 2016), while the credit scores of foreclosed households took years to recover, and some borrowers never recovered at all (Brevoort & Cooper, 2013). The post-manifestation settlement thus continued on the order of 10–15 years. What the emergency compressive capacity of the governance clock handles is the **critical segment of the manifestation function** $M(\tau)$, preventing phase transition from spiraling out of control, not the **post-manifestation long-tail segment**. In this respect it is structurally isomorphic to the pure defense-side intervention in the Japanese case, both unable to shorten the post-manifestation settlement horizon, even though the two intervene in entirely different ways on the $M(\tau)$ segment.

### §1.3 Scenario III: Extreme Separation $\tau_g/\tau_c \approx 10^9$-$10^{14}$

**Data point**: quantitative trading and high-frequency algorithms in China since 2015.

- Execution latency $\tau_c^{exec}$ (latency of quantitative-strategy execution): millisecond to microsecond level, $10^{-3}$-$10^{-6}$ seconds.
- Governance clock $\tau_g$ (regulatory response cycle): measured in months to years, $10^6$-$10^8$ seconds.
- **Clock ratio $\tau_g/\tau_c$: $10^9$ to $10^{14}$.**

This is the **most extreme clock separation** in any known human-governance scenario, 8–13 orders of magnitude larger than the normal accumulation window.

**MCAF implication**: when the clock ratio $\tau_g/\tau_c$ becomes extreme (exceeding $10^6$), **item-by-item manual auditing becomes physically incapable of catching up with the projection side**. Regulators cannot audit millisecond-level operations at millisecond speed, not because the technology is insufficiently advanced, but because the individual's conscious-processing throughput ($\approx$ 10 bits/s) is a biological constant, and the response latency of auditing algorithms is likewise constrained by the physical limits of their deployment chain. Governance must therefore turn to **ex-ante structural constraints and automated execution**—per the mechanism of Chapter 3 §3.1, institutional intervention does not pass through the individual's processing ceiling but directly modifies projection-side variables through the exercise of power: access restrictions truncate unsealed inventory, risk limits and abnormal-order interception compress the per-shot projection load, and necessary rate limits constrain deployment speed (the financial counterpart of "precision speed-limiting" in Chapter 8 §7). Rate limitation is one tool among these, not the only path.

### §1.4 Scenario IV: The Cross-Border Arbitrage Window

The **uneven distribution** of cross-border $\tau_g$ creates a spatial shadow-debt accumulation window. Differences in regulatory density across jurisdictions allow cross-border institutions to exploit the nonsynchronous distribution of $\tau_g$ for regulatory arbitrage. Typical data are as follows: the landing of Dodd-Frank rules in the United States, 2010–2015, preceded the EU's MiFID II in 2018 by 3–8 years; selective adoption in offshore financial centers added another 2–8 years of window. European banks' exposure to U.S. subprime CDOs before 2008 is the typical case of the cross-border arbitrage window: product design was completed in the United States, holdings were dispersed into Europe, and exposure was diffusely distributed through cross-border channels in a way that could not be traced back. The cross-border arbitrage window is the spatial mirror image of the normal accumulation window. It does not receive a separate diagnostic section; this discussion has already been developed in Chapter 19, "The Spatialization of Mapping Chains."

---

**Structural summary of the four clock scenarios.** The normal accumulation window, $\tau_g/\tau_c \approx 10^{0.5}$-$10^{1}$, is the default window for shadow-debt accumulation. Crisis compression mode, $\tau_g \approx 1$ day, is the emergency-response mode unique to finance. Extreme clock separation, $\tau_g/\tau_c > 10^{6}$, makes ex-ante structural constraints the main path. The cross-border arbitrage window is the spatial mirror image of the normal accumulation window. These four clock scenarios coexist in parallel within the same financial system.

MCAF's first structural contribution to financial history lies precisely here: it reorganizes the history of financial institutional evolution from a pulsed narrative of "crisis - legislation - crisis again" into a **typology of the distribution of clock ratios across financial subdomains and historical periods**. This reorganization makes it possible to establish the comparability among the cases addressed later in the three diagnoses of §3: §3.1 on the critical behavior of the manifestation function, §3.2 on the impossibility of pure defense, and §3.3 on the feasibility of structural intervention corresponds to completing stock adjustment through sustained governance **before** manifestation occurs—what it seeks to avoid is precisely the passive slide into crisis compression. The governance implications of extreme clock separation, clock ratio $> 10^{6}$, namely the financial counterpart of the precision speed-limiting path, appear in Chapter 8, §7.

---

## §2 The 258-Year Evolution of the Compensation Loop in the U.S. Financial Market

**Definition of the compensation loop.** The compensation loop of the financial system is a complete loss-processing chain: a risk event occurs -> the loss is recognized -> the loss is allocated by rule among the responsible party, the insurer, the industry fund, and public finance -> the injured party receives at least partial compensation -> the responsible party bears legal consequences. Every node in this chain is supported by one or more institutions: deposit insurance, exchange clearing funds, reinsurance chains, orderly liquidation procedures in bankruptcy law, and the central bank's lender-of-last-resort function. The compensation loop is the core skeleton of the institutional cognitive-debt processing mechanism in finance, and also the very structure whose total absence in the AI domain was diagnosed in Chapter 15, §3—AI lacking an institutionalized chain for converting shadow debt into observable losses (disclosure, accountability, compensation, and related links).

### §2.1 The Prototype of the Compensation Loop in Insurance (1752–1906)

The earliest starting point of the financial compensation loop lies not in banking but in **insurance**, and this fact itself has structural significance: the first mature institutional form through which uncertainty was financially processed was contractual risk-sharing, not ex post loss rescue.

**1752: The Philadelphia Contributionship.** Benjamin Franklin and members of the Philadelphia community jointly founded North America's first continuously operating fire-insurance company, the Philadelphia Contributionship for the Insurance of Houses from Loss by Fire. The core of its institutional design was **mutual loss-sharing**: policyholders jointly contributed to form a pool for the compensation of losses, and when a fire occurred that pool paid the damaged household according to contract. In the North American context, the innovation of this mechanism did not lie in compensation itself, since guild-style mutual aid already existed in ancient Rome and contractual fire insurance had appeared in seventeenth-century London, but in the **institutionalization of contractual obligation**: compensation no longer depended on the charitable goodwill of any government or individual, but became the legally defined right and obligation of a continuously operating organization. At this stage, the compensation loop was still a mechanism **endogenous to a single industry**: a fire-insurance company compensated fire losses, while risk types outside the contract remained outside the compensation chain.

**1871: The Great Chicago Fire and the reinsurance network.** The Great Chicago Fire of 1871 destroyed about 17,450 buildings (Payan, 2013); 58 insurance companies went bankrupt as a result, and the industry ultimately paid only about half of policyholders' claims (less than one-third of the total fire loss) (*Encyclopedia of Chicago*), exposing the inability of a single-layer compensation structure to bear tail risk. The event drove the rapid expansion of reinsurance markets, Lloyd's, Munich Re, Swiss Re, and the compensation loop expanded from a single-layer contract into a **multilevel cascading** structure.

**1906: The San Francisco earthquake.** The loss was about $350–500 million, about 1.3%-1.8% of U.S. gross national product (GNP) at the time (Odell & Weidenmier, 2004). Although some British insurance companies initially invoked the "earthquake exclusion" clause in their policies to refuse payment, under the strong protests of the San Francisco public and the pressure of legal opinion, most companies eventually paid in full under the "fire" clause (Odell & Weidenmier, 2004; Fradkin, 2005). That contractual obligations were, after such bargaining, still confirmed and fulfilled under extreme conditions confirmed the enforceability of insurance contracts under extreme loss: the status of insurance as a calculable, priceable, and tradable financial instrument received institutional confirmation through this stress test.

**The three structural features of the insurance prototype of the compensation loop** are these. First, the compensation loop is endogenous to contract and does not depend on ex post government rescue. Second, risk dispersion is achieved through reinsurance, breaking through the capacity limit of a single-layer pool by means of multilevel cascading. Third, contractual obligations are still fulfilled under extreme events, which is the core source of insurance's legitimacy as a financial instrument. These three features were gradually transplanted in the twentieth century into banking and securities, but every such transplantation came at the cost of a crisis.

### §2.2 Crisis-Driven Construction of the Compensation Loop in Banking (1929–1963)

**1929–1933: The Great Depression and the creation of the FDIC.** After the stock-market crash in October 1929, the United States entered the Great Depression. Before 1933 the United States lacked a broad, stable, fed-

eral deposit-insurance system; the earlier state-level arrangements had largely failed and could not stem the systemic runs of the Great Depression. More than 9,000 banks suspended operations between 1930 and 1933 (1,350, 2,293, 1,453, and 4,000 respectively), and depositors' losses totaled about $1.3 billion at then-current values (FDIC, 1984). When a bank suspended operations, depositors could only participate in bankruptcy liquidation as general creditors, with actual recovery rates very low and very slow. Bank runs became the self-amplifying mechanism of banking crisis: rumor alone that one bank was in trouble could trigger a run by depositors; the run then amplified the bank's problems and made it truly fail; that failure then triggered runs on other banks. This is the prototype, eighty years earlier, of the money-market-fund run in the 2008 Sorkin narrative, structurally isomorphic across the interval.

On June 16, 1933, Franklin D. Roosevelt signed the Banking Act of 1933, the Glass-Steagall Act, and the Federal Deposit Insurance Corporation, FDIC, was thereby established. The institutional design of the FDIC was **the transplantation of the contractual compensation prototype of insurance into bank deposits**: deposit-taking banks paid insurance premiums to the FDIC, and the FDIC provided compensation guarantees for depositors when banks failed. The coverage ceiling was adjusted several times from $2,500 in 1934 and was permanently fixed at $250,000 by the Dodd-Frank Act in 2010.

The creation of the FDIC was the **first institutional achievement** in transplanting the compensation loop from insurance into banking. This transplantation took about 180 years, from the Philadelphia Contributionship in 1752 to the FDIC in 1933, and this duration is itself a data point for the governance clock $\tau_g$ in the sense of institutional creation within the MCAF framework. The two observation nodes of 1752 and 1933 are 181 years apart—this span reflects the historical evolution path of the relevant institutions in the United States and is not the required duration of cross-industry transplantation; later fields can borrow existing legal, insurance, and organizational technology, and their construction cycles must be judged separately.

**1963: The Haupt event and the beginning of industry's spontaneous compensation, driven by a mix of reputational and legal pressure.** Ira Haupt & Co. collapsed because of the "salad oil scandal," putting the funds of about 20,700 clients at risk. The New York Stock Exchange (NYSE) announced that it would protect client accounts with up to $12 million; the final amount actually paid, apportioned among member firms, was about $9.5 million, completing the return of clients' securities and funds (Seligman, 1982; Miller, 1965). American Express, drawn into the same scandal in the same period, had a warehousing subsidiary that issued bogus warehouse receipts against about $100 million of bank loans; facing creditors' claims and the exposure of the parent company's legal liability, in April 1964 Amex proposed a settlement, an upfront payment of $35 million plus $10 million in installments and a guarantee of insurance recoveries, totaling about sixty percent of the claims, declaring that the move would "strengthen the good name of American Express and its subsidiaries" (Miller, 1965). The Haupt event directly drove the establishment of the NYSE Special Trust Fund in 1964 and the Securities Investor Protection Corporation (SIPC, with the current compensation ceiling of $500,000 per customer), established in 1970. These two events established a **soft layer of the compensation loop, driven by a mix of reputational and legal pressure**, beyond legal compulsion: the NYSE's compensation was not driven by legal obligation, while Amex's settlement was jointly propelled by reputation maintenance and the legal-liability risk of the parent company.

### §2.3 The Dilemma of the Compensation Loop under Systemic Risk (1998–2008)

The compensation loops of insurance and banking were basically built between the 1930s and the 1970s, but two structural problems remained unresolved. The first was the compensation of **cross-industry systemic risk**: when losses crossed traditional industry boundaries, industry-specific compensation loops could not cover them. The second was compensation for **ultra-large-scale losses**: when the loss from a single event exceeded the combined capacity of all industry compensation pools, who stood behind the tail end of the loop?

**1998: LTCM and the first manifestation of cross-industry systemic risk.** Long-Term Capital Management, founded in 1994 by Meriwether, Merton, and Scholes, used about $4.7 billion in capital to lever positions of $125 billion and a notional exposure of more than $1 trillion (Lowenstein, 2000). Russia's sovereign default in August 1998 triggered a global flight to safety, and LTCM's capital shrank from $4.7 billion to about $560 million. Its counterparties covered nearly every major investment bank, and disorderly bankruptcy liquidation would have triggered cascading fire sales. The president of the New York Fed, McDonough, coordinated 14 institutions to inject $3.625 billion so that LTCM could be unwound in an orderly manner—the first cross-sub-industry **private rescue coordination** in the modern financial system to include a hedge fund (Edwards, 1999). The structural feature was that the Federal Reserve provided the coordination platform, procedural legitimacy $L_{proc}$, but did not contribute capital; the beneficiaries and the contributors overlapped only in part. MacKenzie (2006) tracks how LTCM's strategies were widely imitated to form a "superportfolio," the first precisely documented case of "strategy homogenization -> liquidity collapse," structurally isomorphic to the CDO market in 2008.

**2008: The financial crisis and the full stress test of the compensation loop.** The details of the 2008 financial crisis are unfolded in §3.1; here only its location in the evolutionary history of the compensation loop is marked. During the crisis, every layer of the compensation loop was triggered simultaneously: FDIC compensation for bank deposits, as multiple banks failed in 2008; SIPC compensation for broker customers after Lehman's collapse; insurance and reinsurance chains under stress during the crisis, AIG's crisis, however, arising from securities lending and credit-derivative business rather than the traditional underwriting chain; central-bank intervention through two distinct sets of instruments, emergency lending as lender of last resort and massive asset purchases; the $700 billion legislative authorization for TARP; and the Federal Reserve's $85 billion secured credit line to AIG, with total government support later reaching about $182 billion.

The key structural feature of the 2008 compensation loop was this: when all marketized compensation mechanisms were exhausted, **sovereign credit functioned as the ultimate reinsurer**. The combined $85–182 billion of government support provided when AIG collapsed (the Federal Reserve's secured credit line and subsequent Treasury funds) did not come from any pre-funded compensation pool; it was the final obligation borne by taxpayers through sovereign credit. This layer of compensation completed the full chain of the compensation loop from **industry spontaneity** -> **statutory institution** -> **cross-industry systemic** -> **sovereign ultimate**. The 45 years of institutional evolution between Haupt in 1963 and AIG in 2008 ultimately established sovereign credit as the final backstop of the compensation loop.

**2010: The Dodd-Frank Act and the orderly liquidation mechanism.** The political cost of the 2008 rescue was immense, and TARP's initial rejection in the House immediately reflected public resistance to "taxpayers paying for Wall Street." The post-crisis legislative response was not to cancel the sovereign backstop, since that is practically irretractable in a real crisis, but to establish an **Orderly Liquidation Authority (OLA)**. Title II of the Dodd-Frank Act granted the FDIC takeover and liquidation powers over systemically important financial

institutions.

The institutional design of OLA attempted to resolve the dilemma of "too big to fail": allowing large financial institutions to fail while preventing cascading collapse through an orderly process. The Dodd-Frank Act addresses the exit of large institutions through two interlocking arrangements: Title I requires systemically important institutions to submit resolution plans ("living wills") detailing how they would be orderly broken up in the event of failure; Title II establishes the OLA to provide a special resolution path when ordinary bankruptcy is insufficient to preserve financial stability—its activation requires a statutory recommendation by the Federal Reserve and the FDIC and a determination by the Secretary of the Treasury in consultation with the President, with the FDIC executing the takeover and allocating losses to creditors according to a preset hierarchy; an orderly-liquidation fund provides transitional liquidity support. OLA is not the cancellation of the "too big to fail" narrative, but a **structural rewriting of that narrative**: from "too big to fail, therefore government must rescue" to "too big to fail chaotically, therefore it must be orderly liquidated."

### §2.4 The Evolution of the Compensation Loop in the Chinese Context (2015–2026)

The evolution of the compensation loop in China has been compressed by about 80 years relative to the U.S. path: from the stripping of non-performing assets from state-owned banks in 1998 to the normalization of net-value management for wealth-management products in 2026, roughly 28 years complete a path of institutional evolution that took the United States about 80 years from 1930 to 2010. This compression is a feature of **latecomer institutional importation**: China can selectively borrow from U.S. institutional design and avoid some of its detours, for example by beginning directly with a modern FDIC-style deposit-insurance scheme instead of replaying the bank runs of the 1930s under a regime without deposit insurance. But it also bears the cost of lateness: the asynchronous development of financial innovation and regulatory capacity caused regulatory look-through capacity to lag behind when shadow banking expanded rapidly in the 2010s.

**2015: The establishment of the Deposit Insurance Regulation.** China implemented the Deposit Insurance Regulation on May 1, 2015, and the deposit-insurance fund was established under the leadership of the People's Bank of China. The compensation ceiling was RMB 500,000 per depositor. In MCAF terms, deposit insurance is the minimum procedural device of the legitimacy base $\|L\|$ in banking. Before it, the de facto compensation of bank deposits relied on implicit government guarantees. In normal times those implicit guarantees were equivalent to, and sometimes stronger than, the 500,000 ceiling; in extreme crisis, however, they could become unstable precisely because of the lack of explicit rules. The core structural contribution of the Deposit Insurance Regulation is not that it raises the level of compensation in normal times, but that it **transforms implicit guarantees into explicit contractual rights**, thereby reducing ambiguity in the legitimacy base under extreme scenarios.

**2018–2021: The New Asset Management Regulations and the structural compression of shadow banking.** The issuance in 2018 of the Guiding Opinions on Regulating the Asset Management Business of Financial Institutions, the New Asset Management Regulations, was China's structural intervention in the shadow-banking domain. Its concrete contents and MCAF mapping are unfolded in §3.3. Here only its location in the evolutionary history of the compensation loop is marked: the core of the New Asset Management Regulations was **to break implicit-guarantee expectations**, requiring wealth-management products to be valued and displayed on a net-value basis, so that investors directly face the fluctuation of underlying assets. This operation shifted shadow debt that had previously been borne outside the compensation loop through "implicit bank back-

stopping" back onto investors themselves. **Explicit compensation boundaries** replaced **implicit unlimited guarantees**.

This was a major structural change to the compensation loop, but its direction was the opposite of the creation of the FDIC in 1933: rather than expanding compensation coverage by creating a new compensation mechanism, it **shrank the outer perimeter of the compensation loop back to calculable contractual coverage by clarifying the boundaries of non-compensation**. In the MCAF framework, the common point of the two directions is that both convert implicit expectations into explicit rules; their effects on narrative load $\phi$ run in opposite directions—breaking implicit guarantees depresses the principal-protection narrative, while deposit insurance elevates the safety narrative of insured deposits—this chapter treats them as two faces of one operation in the sense of legitimacy structure rather than monotonicity of $\phi$, both converting implicit shadow debt into explicit observable debt while raising the clarity of meta-audit $A$.

**2019: The takeover of Baoshang Bank and the breaking of the narrative that "big banks do not fail."** On May 24, 2019, the People's Bank of China and the China Banking and Insurance Regulatory Commission jointly announced the takeover of Baoshang Bank, the first bank-takeover event in China in twenty years. The structural significance of this event is that, following the closure of Hainan Development Bank in 1998, it once again confirmed through a market-based, law-based disposal path that commercial banks can exit the market, while protecting individual depositors through the deposit-insurance fund and public funds and applying differentiated arrangements to institutional claims. The point of comparison with the Continental Illinois event of 1984 in the United States is that both handled systemically important institutions with the combination of "institutional risk disposal + full protection of specific claims," but in opposite directions—Continental Illinois thereby established "too big to fail" (general creditors were also fully protected), whereas Baoshang for the first time imposed explicit losses on institutional claims.

In the Baoshang case, all individual deposits were fully protected in principal and interest (the portion above the RMB 500,000 ceiling of the Deposit Insurance Regulation being backstopped by additional funds from the deposit-insurance fund and the People's Bank of China); corporate deposits and interbank liabilities of RMB 50 million or less were fully protected in principal and interest, while the portion above RMB 50 million was handled through equal negotiation between the takeover group and creditors and protected according to law; according to the tiered repayment scheme disclosed by the market, interbank liabilities of RMB 100 million to 2 billion received ninety percent of principal, with the repayment ratio declining as the scale increased (21st Century Business Herald, 2019). This differentiated treatment of "zero loss for individuals, partial loss for institutions" turned breaking rigid guarantees into an observable practical sample at the level of institutional claims for the first time.

**2020–2026: Real-estate debt restructuring and the progressive construction of the compensation loop.** After 2020, debt problems in the real-estate sector became another major node in the evolution of China's compensation loop. In the second half of 2020 China began applying the "three red lines" regulatory framework to the real-estate sector; the Evergrande event in 2021 triggered concerns about systemic default across the entire property-developer sector; from August 2022 onward, a special-purpose "ensuring housing delivery" loan with an initial scale of RMB 200 billion was established through policy banks (People's Daily Online, 2022; Caixin, 2022), placing the completion funding of presold but undelivered projects at the highest priority within the compensation loop; major developers such as Evergrande, Country Garden, and Sunac proceeded through debt restructuring, split treatment of domestic and offshore debt, and asset sales; wealth-management products

experienced two rounds of net-value breakage in 2022, one in spring and one in autumn; and commercial-housing sales area shrank from a peak of about 1.8 billion square meters in 2021 to about 880 million square meters in 2025 (National Bureau of Statistics, annual Statistical Communiques on National Economic and Social Development). These events together form a compensation-loop adjustment process lasting about six years through 2026.

Its characteristic is that some disposal arrangements accorded policy-level priority to individual depositors and the delivery of presold housing—but this does not constitute a unified repayment hierarchy applicable to all assets and all procedures; the concrete order still depends on the distinction among deposits, purchase claims, wealth-management investments, secured claims, and equity (statutory repayment order following bankruptcy law), while the orientation of prioritizing housing delivery, protection of residential rights, over wealth-management repayment, protection of financial rights, with equity claims last, recurred across multiple events.

**Structural difference between the Chinese and U.S. paths.** The typical pattern of U.S. compensation-loop evolution is a pulsed cycle of "crisis -> legislation -> deregulation -> crisis again": Panic of 1907 -> Federal Reserve System in 1913 -> deregulation in the 1920s -> Great Depression -> Glass-Steagall in 1933 -> deregulation in the 1980s, relaxation of Regulation Q and partial repeal of Glass-Steagall -> 2008 crisis -> Dodd-Frank in 2010 -> partial relaxation after 2018. China's compensation-loop evolution adopts a pattern of **proactive construction + stress testing + progressive optimization**: deposit insurance in 2015 -> New Asset Management Regulations in 2018 -> the Baoshang takeover in 2019 as a stress test -> progressive treatment of real-estate debt from 2020 to 2026. In the MCAF framework, these two paths correspond to different slope shapes of the manifestation function $M(\tau)$: the United States is pulsed, the slope of $M(\tau)$ becomes extremely steep in crisis, and post-crisis relaxation allows shadow debt to reaccumulate; China is progressive, the slope of $M(\tau)$ is deliberately flattened, but the accumulation duration is lengthened. How the total costs of the two paths compare is not a question this chapter seeks to answer; here it only marks the difference and leaves the discussion to later chapters and future empirical research.

### §2.5 The Three-Layer Structure of the Compensation Loop and the MCAF Conclusion

The 258-year evolution has produced a **three-layer concentric structure** of the financial compensation loop, each layer corresponding to one kind of compensation trigger and one source of funds. The three-layer structure below takes the U.S. model as typical; under other institutional environments the concrete carriers of the three layers may differ (for example, in China the third layer manifests mainly as implicit land finance and the assumption of local-government debt, rather than quantitative easing (QE)).

**First layer: contractual compensation.** Its core mechanisms include insurance contracts, deposit insurance, SIPC, reinsurance chains, and priority-repayment rules in bankruptcy law. Its characteristics are explicit compensation ceilings, independent funding pools, and contractual trigger conditions. The first layer is the default state of the compensation loop and independently bears compensation responsibility in the overwhelming majority of financial-loss events. In MCAF terms, the triggering of first-layer compensation corresponds to the **institutionalized routine conversion** of the manifestation function $M(\tau)$ from shadow debt $D_c^{shadow}$ to observable debt $D_c^{obs}$: the rules are clear and the path is predictable.

**Second layer: systemic rescue.** Its core mechanisms include the central bank as lender of last resort, TARP-

style government capital injection, and orderly liquidation of systemically important institutions. Its characteristics are blurred compensation ceilings, sovereign credit as the source of funds, and the trigger condition of "preventing systemic collapse." The second layer is the emergency state of the compensation loop, activated when the capacity of the first layer is exhausted. In MCAF terms, the triggering of second-layer compensation corresponds to the **critical phase-transition segment** of the manifestation function $M(\tau)$, where standard contractual mechanisms fail and emergency mobilization of the legitimacy base $\|L\|$ must substitute for the routine path.

**Third layer: socialization of loss.** Its core mechanisms include QE, sovereign-debt expansion, and inflationary dilution. Its characteristics are no explicit compensation ceiling, costs borne by the whole population rather than any specific responsible party, and legitimacy that usually rests on ex post political ratification rather than a preexisting contract for that specific loss (the central bank's policy instruments themselves are separately authorized by statute). The third layer is the terminal backstop of the compensation loop, activated when the first and second layers are exhausted but system losses remain undigested. In MCAF terms, third-layer compensation corresponds to the payout of costs in the **post-manifestation long-tail segment**: losses do not manifest through discrete jumps in the manifestation function, but are amortized over the long term through **continuous social processes** such as inflation, dilution of sovereign credit, and intergenerational wealth transfer. This layer is in essence not "compensation" at all, but the pushing of uncleared debt farther out along the time dimension.

The three layers differ in whether responsibility is predetermined, whether funds are prefunded, and who bears the cost. The three layers are not strictly sequential in activation: contractual compensation, systemic rescue, and macroeconomic adjustment may occur simultaneously and their functions may overlap—the purpose of the layering is to distinguish responsibility boundaries and cost-bearing arrangements, not to set an activation sequence. In the 2008 crisis, all three layers were triggered **simultaneously**, completing the first full-chain stress test of the compensation loop: first layer, FDIC compensation for depositors of failed banks; second layer, TARP rescue of systemically important institutions such as AIG; third layer, the long-run distributional consequences of the Federal Reserve's massive asset purchases and other macro policies—changes in asset prices and the public-debt path, whose inflationary effects remain contested in the literature. The combined cost of these three constituted the complete record of the 2008 crisis within the compensation loop.

**MCAF conclusion.** Every advance in the 258-year evolution of the compensation loop corresponds to a structural stress event of the compensation loop—the portion of these events in which risk was underestimated, responsibility was ambiguous, or preparation was insufficient, and undigested burdens were thereby deposited, is what enters this chapter's analysis of cognitive debt and the manifestation function $M(\tau)$. The Great Chicago Fire of 1871 corresponds to the first structural stress event inside the insurance industry, where the capacity of the single-layer compensation pool was breached; the Great Depression of 1929–1933 corresponds to the first structural stress event in banking, namely systemwide bank runs; the Haupt event of 1963 corresponds to the first structural stress event in securities, namely the establishment of industry's spontaneous compensation responsibility; the LTCM event of 1998 corresponds to the first critical $M(\tau)$ event of cross-industry systemic risk, namely the activation of a private-rescue coordination mechanism; and the 2008 financial crisis corresponds to the simultaneous activation of all three layers of the compensation loop. Every stress event pushes forward the next round of institutional construction, forming an iterative cycle of **crisis -> manifestation -> institutionalization**. This observation is the historical basis of §3.1 below: the critical behavior of $M(\tau)$ is not an isolated case in 2008, but the **regular driving mechanism** of financial institutional evolution.

At the same time, these 258 years of evolution provide two orders-of-magnitude references for the diagnosis in Chapter 15, §3 that the AI domain suffers from a "missing manifestation chain." First, the cycle of institutional construction from industry prototype to cross-industry transplantation is measured in **centuries** (a reference anchored to the single U.S. path, not a minimum construction period for AI). Second, the total cycle from single-layer compensation to fully developed three-layer structure is measured in **250+ years**. The AI domain is currently still at the prehistory stage of compensation-loop evolution: sporadic contractual indemnity commitments by vendors, such as the copyright indemnity clauses of generative-AI services (Microsoft, 2023), and individual insurance products exist, but no cross-agent, standardized, independently funded first layer of contractual compensation has yet formed, let alone an industry-level mutual-sharing prototype. This order-of-magnitude reference is the most direct contribution of this chapter to the diagnosis in Chapter 15.

---

## §3 Three MCAF Diagnoses

This section unfolds the three structural diagnoses that this chapter extracts from the MCAF perspective. A preliminary declaration is needed here: all structural assertions in this chapter are confined to the range of observed cases, and cross-institutional generalizability requires testing on a larger sample. Each diagnosis points to one historical case as its primary evidence and concludes with a falsifiability condition. The criterion for selecting these three diagnoses has already been stated in the methodological section: each diagnosis must pass the test—if the conclusion still holds once the MCAF variables are removed, then the diagnosis will not be unfolded here. Nor does this section perform a complete six-variable mapping of each case, because that would make the chapter expand without limit and become highly redundant with the existing financial literature. What is retained here is only the structural increment that MCAF can contribute.

**The logical relation among the three diagnoses.** The three diagnoses address three temporal positions in financial governance. Diagnosis I, critical phase transition, answers the question "what happens after the accumulation window lasts too long" -> nonlinear jump of $M(\tau)$. Diagnosis II, the impossibility of pure defense, answers "what happens if only the defense side is intervened upon after manifestation" -> the post-manifestation settlement lengthens to 14–25 years. Diagnosis III, progressive governance, answers "can proactive intervention before manifestation avoid phase transition" -> yes, but only with sustained governance plus multivariable synchronization. Together they provide complete coverage of **pre-manifestation / mid-manifestation / post-manifestation**, with intervention dimensions corresponding respectively to the projection side, $\Lambda/\phi$, emergency response, compression of $\tau_g$, and the defense side, $A/\|L\|$. Japan chose "post-manifestation + pure defense side"; the United States chose "mid-manifestation + emergency response"; China chose "pre-manifestation + intervention on the projection side."

### §3.1 Diagnosis I: The Critical Phase-Transition Behavior of the Manifestation Function $M(\tau)$ (Primary Evidence: United States, 2008)

**Diagnostic proposition**: when the clock ratio $\tau_g/\tau_c$ remains for a long time within the normal accumulation window, about $10^{0.5}$-$10^{1}$, meaning that the regulatory cycle is 3–10 times the market-doubling time, shadow debt $D_c^{shadow}$ accumulates slowly within it. Once some exogenous shock, decline in housing prices, default of a key institution, or repricing of a key asset class, triggers the conversion of shadow debt into explicit debt, the manifestation function $M(\tau)$ exhibits a **nonlinear critical phase transition**: within a very short time, on

the order of weeks, it jumps from a gentle rise to a steep climb; what fails is the capacity of standard contractual mechanisms and regulatory audit to preserve system stability under the prevailing conditions, not the cessation of those mechanisms themselves—margin calls, revaluation, and redemption were in fact executing compulsorily and amplifying the pressure; and the governance clock $\tau_g$ must be compressed from its normal scale of years to the scale of days in order to prevent systemic collapse. This phase transition is not the result of "regulators reacting too slowly," but the **inherent structural behavior of shadow debt once accumulated to the critical point**. Even if regulatory procedures operate lawfully and normally, insufficiently identified correlated exposures may still be concentratedly revealed after a trigger event (the 2008 case supports this nonlinear manifestation mechanism, but this chapter does not thereby assert that any clock ratio necessarily triggers a crisis).

This is what MCAF can say about the 2008 crisis that existing analyses cannot. Existing analyses, Brunnermeier 2009, Gorton 2010, Sufi & Taylor 2022, explain at the mechanism level the failure of CDO pricing models, the run dynamics of shadow banking, and the procyclicality of financial-institution leverage. These analyses are structurally correct, but their narrative form is "how some chain of mechanism failed in 2008," a causal narrative. The increment of MCAF is to reorganize these mechanism chains as **the shape of one manifestation function on the time axis**, and to point out that this shape is the inherent structural consequence of an accumulation window generated by $\tau_g/\tau_c \approx 10^{0.5}$-$10^{1}$, rather than a contingent narrative specific to 2008.

### §3.1.1 Definition and Measurement of the Manifestation Function $M(\tau)$

The manifestation function $M(\tau)$ is defined as the proportion of accumulated shadow debt that has been converted into explicit debt by time $\tau$ from some time baseline. Under the authorization of Chapter 3 §4.2 (the domain, range, and concrete form of the manifestation function are all given by the scenario model), this chapter concretizes it in the financial scenario as a cohort-based ratio form—taking as the object of observation the same batch of debt already formed at the base time point $t_0$, whose ex post estimated total is denoted $D_c^{true}(t_0)$ and whose cumulative confirmed amount by $\tau$ is denoted $D_c^{obs,cum}(\tau \mid t_0)$:

$$M(\tau \mid t_0) = \frac{D_c^{obs,cum}(\tau \mid t_0)}{D_c^{true}(t_0)}$$

Amounts already paid or written off are not deducted from the cumulative confirmed amount, and newly formed debt is not merged into the cohort; $M$ describes the process of loss recognition, not the process of loss settlement.

The value of $M(\tau)$ varies within the interval $[0, 1]$. $M(\tau) = 0$ means that all cognitive debt remains in the shadow layer, $D_c^{obs} = 0$; $M(\tau) = 1$ means that all cognitive debt has completely manifested into observable institutional records. The **slope** of $M(\tau)$, $dM/d\tau$, describes the speed of manifestation. **Methodological limitation**: the true debt $D_c^{true}(t_0)$ under the cohort-based caliber can only be estimated ex post (shadow debt is by definition debt that has not yet been observed), so the absolute value of $M(\tau)$ cannot be directly measured. It can only be indirectly judged through proxy time series such as CDS spreads, rating-downgrade speed, or NPL ratios, so as to identify its **shape characteristics**, linear vs. nonlinear jump. The use of $M(\tau)$ in this chapter focuses on this shape characteristic and does not claim to provide point estimates. During the normal periods of the compensation loop discussed in §2, for most of the 1970s through the 1990s, $dM/d\tau$ remains low, reflecting the routine operation of first-layer contractual compensation. In the critical-event periods of the compensation loop, 1871, 1929, 1998, and 2008, $dM/d\tau$ jumps to a high level.

The manifestation function $M(\tau)$ can be indirectly estimated in the financial scenario, via the proxy variables below, as to its **shape characteristics** (with the methodological limitation of the preceding paragraph unchanged); this chapter does not provide point estimates. Concrete empirical proxies include the non-performing-loan ratio, CDS spreads, rating-downgrade speed, the scale of margin calls, and the usage rate of central-bank liquidity tools. These proxies have complete daily, and sometimes hourly, records for the 2008 crisis, making it possible to reconstruct the shape of $M(\tau)$ ex post.

### §3.1.2 September 13, 2008: Sorkin's Narrative as Structural Evidence

September 13, 2008, a Saturday, was the key time point of the phase transition of the manifestation function $M(\tau)$. Sorkin's (2009, *Too Big to Fail*) record of the weekend CEO meeting in the New York Fed building provides interview- and document-based reconstructed narrative evidence of the phase transition of $M(\tau)$. Its structural feature is the **separation between internal sight and public sight**:

- **Internal sight**, inside the Fed building on September 13–14: participants had already, on the basis of internal due-diligence materials, come to suspect that Lehman's true leverage far exceeded what its public reports showed (a suspicion formally confirmed two years later by the Valukas Report (2010) through the "Repo 105" accounting maneuver); that AIG's CDS exposure would be triggered; and that major exposure of money-market funds to Lehman commercial paper would trigger a run.
- **Public sight**, market signals at the same time: Lehman CDS spreads closed below 600 basis points on September 10 and widened to 775–800 basis points in early trading on September 11 (MarketWatch, 2008), while the three major rating agencies still assigned A/A+ ratings. In public sight, $M(\tau)$ still remained low.

This separation is precisely the typical form before the critical phase transition of $M(\tau)$: internal sight already strongly suspected that the scale of shadow debt far exceeded the explicit record, while public sight still displays the contractual mechanism as operating normally.

### §3.1.3 September 15–16: The Jump in the Slope of the Manifestation Function $M(\tau)$

On the morning of Monday, September 15, Lehman formally filed for Chapter 11 bankruptcy protection. The sequence of events that day occupies the following place in the phase transition of $M(\tau)$:

- **Before Monday's opening**: news of Lehman's bankruptcy becomes public. Internal sight aligns with public sight for the first time.
- **On Monday**: the Dow Jones falls 504 points, 4.4%; Lehman's stock drops from about $3.65 to about $0.21; AIG shares fall 65%, closing at $4.76 (Sorkin, 2009; Valukas, 2010); Washington Mutual, Merrill Lynch, and other institutions fall in tandem (Merrill's decline remaining relatively limited because of its already-concluded merger agreement with Bank of America).
- **Monday night**: AIG submits an emergency liquidity request to the Federal Reserve; the newly discovered, roughly $20 billion of unrecorded losses in its securities-lending business found over the weekend by JPMorgan's due-diligence team further widened the funding gap (the business's total eventual losses exceeded $20 billion, McDonald & Paulson, 2014), and the credit facility the Fed finalized the next day was $85 billion (Sorkin, 2009).
- **Tuesday, September 16**: the Reserve Primary Fund, a money-market fund, announces that its holdings

of Lehman commercial paper have pushed its NAV below \$1, "breaking the buck," triggering a run across the money-market-fund industry: institutional investors withdrew more than \$172 billion from money-market funds within one week of the default (Kacperczyk & Schnabl, 2013), and total redemptions from prime money-market funds over the full week (September 15–19) were about \$300 billion (ICI, 2009). A more granular event study further shows that the run was concentrated in institutional funds: Schmidt, Timmermann & Wermers (2016) measure cumulative outflows from prime institutional money-market funds of about \$400 billion in the first two weeks after Lehman's bankruptcy (this figure is on a "prime institutional only / first two weeks / cumulative outflow" basis, which differs from the ICI "all prime / one week / net redemption" basis above, and the two sets of numbers must not be conflated).

- **Tuesday night**: the Federal Reserve announces a secured loan of up to \$85 billion to AIG, taking a 79.9% equity participation.
- **Friday, September 19**: the Treasury announces a temporary guarantee for money-market funds.

From September 13, when internal sight first became fully clear, to September 19, when the guarantee program for money-market funds was announced, **the manifestation function $M(\tau)$ completed a nonlinear jump from low level to high level within a few days**—this dense sequence supports the judgment of concentrated risk manifestation, but does not warrant the claim that $M(\tau)$ rose from some low value to some high value within a specific number of days. This jump was not the result of regulators "speeding up their reaction"; throughout the entire process they remained in a passively chasing position. Rather, it was the **automatic forced manifestation of shadow debt itself once the trigger event occurred**: Lehman's bankruptcy automatically triggered the repricing of its counterparties' exposures; CDS counterparties were forced to mark losses to market; money-market funds were forced to disclose their commercial-paper exposure; rating agencies were forced to downgrade related entities. This cascading manifestation process does not depend on regulatory intervention. It is the phase-transition behavior of the shadow-debt structure itself at the critical point.

#### §3.1.4 The Paradox of the "Unpredictability" of the Critical Behavior of the Manifestation Function $M(\tau)$

A key MCAF observation is that the **macro-structure** of the phase transition of $M(\tau)$ is predictable, once $\tau_g/\tau_c$ remains within the normal accumulation window for long enough, phase transition will occur sooner or later, but the **precise time point and trigger event** are unpredictable.

Already in 2006–2007 a number of analysts warned about the housing bubble and the fragility of the CDO market: Raghuram Rajan's warning at Jackson Hole in 2005, "Has Financial Development Made the World Riskier?"; Nouriel Roubini's 2006 warning about collapse in the housing market; and Michael Burry's short strategy against subprime CDOs, later recorded in Michael Lewis's *The Big Short*. These warnings were correct at the level of macro-structural judgment: the scale of accumulated shadow debt had indeed moved beyond the visible range of regulation. But none of them could precisely predict the triggering event of the phase transition, Lehman rather than Bear Stearns, September 7 with the takeover of Fannie and Freddie or September 15, nor its exact time point.

In the MCAF framework this paradox is not an anomaly, but an **inherent property** of the phase transition of the manifestation function $M(\tau)$. The structural analogue of the phase transition of $M(\tau)$ is self-organized criticality in physics (Bak, 1996): when the system is near a critical point, the scale of the triggered event cannot be predicted from the external characteristics of the single event; only the cumulative degree of the system as a whole can be predicted. The concrete trigger event is random, but as long as the accumulation window persists,

a phase transition will occur sooner or later—this chapter does not thereby assert that the crisis was bound to fall in the specific year 2008; the trigger event observed ex post, Lehman's bankruptcy on September 15, was only one of several possible paths, and the successive falls of neighboring institutions within weeks (Washington Mutual taken over on September 25, Wachovia sold on September 29) suggest that the system was already in a critical state.

This observation has direct analogical significance for current AI regulation, to be unfolded in §4: the specific trigger event of an AI crisis cannot be predicted, but the current order of magnitude of $\tau_g/\tau_c$ in AI, at least comparable to the normal accumulation window of finance and perhaps more extreme, means that the phase transition of AI's manifestation function $M(\tau)$ is an inherent consequence of the accumulation window, not a matter of whether regulators are "vigilant enough."

### §3.1.5 An MCAF Reading of Greenspan's 2008 Testimony

The testimony given by former Federal Reserve Chairman Alan Greenspan on October 23, 2008 before the U.S. House Committee on Oversight and Government Reform is a crucial first-hand record of the cognitive limitation of regulators before the phase transition of $M(\tau)$. Committee Chairman Henry Waxman pressed Greenspan on whether his ideology had led to regulatory failure, and Greenspan admitted that his model of the world contained a "flaw":

> Chairman WAXMAN. In other words, you found that your view of the world, your ideology, was not right, it was not working. (Greenspan, 2008)
>
> Mr. GREENSPAN. Precisely. That's precisely the reason I was shocked, because I had been going for 40 years or more with very considerable evidence that it was working exceptionally well. (Greenspan, 2008)

Read through the MCAF framework, this testimony means that Greenspan was not admitting a "mistaken judgment," but a **systemic blind spot in meta-audit** $A$. The auditing tools at the macro-audit layer, $A_{macro}$, monetary-policy analysis and financial-stability assessment, were constrained by ideological legitimacy $L_{ideo}$, the ideology of "market self-regulation," and could not independently recognize that the accumulation of $C \cdot \Lambda \cdot \phi$ had already exceeded the defensive capacity of $A \cdot E_c \cdot \|L\|$.

The institutional constraint of $L_{ideo}$ does not exclude local dissent. On May 7, 1998, Brooksley Born, then Chair of the Commodity Futures Trading Commission (CFTC), released the Concept Release Concerning Over-the-Counter Derivatives Market and argued for regulatory supervision of over-the-counter derivatives; the proposal was immediately opposed jointly by then Treasury Secretary Rubin, Deputy Secretary Summers, and Federal Reserve Chairman Greenspan. After the hearing of the House Banking Committee in October 1998 and the appropriations-bill freeze clause, the 1999 Report of the President's Working Group on Financial Markets effectively froze the path toward CFTC expansion of authority, and the Concept Release never entered legislation (CFTC, 1998; Born, 1998; Haggerty, 2022). The significance of this counter-case in MCAF is not to deny the existence of constraint from $L_{ideo}$, but to show that the filtering of meta-audit $A$ by $L_{ideo}$ is not "ideological homogeneity prevents dissenting voices from arising," but rather "dissenting voices cannot break through the collective legitimacy filter at the meso layer and enter the policy path."

The significance of Greenspan's testimony does not lie in "an important regulator admitted error," but in confirming that **regulators' cognitive capacity is itself constrained by the three dimensions of the legitimacy**

**base** $\|L\|$. When ideological legitimacy $L_{ideo}$ is deeply internalized as "market self-regulation," the set of auditing tools at the meta-audit layer $A$ is prefiltered so that the possibility of "systemic market failure" is excluded in advance. This filter is not a matter of personal bias, but an institutional constraint. Those serving as Federal Reserve Chair in that generation's ideological environment were prone to forming similar blind spots; Born's dissent shows that the institutional environment was not without divergent judgments—what needs explanation is why such judgments failed to enter the policy path. This is institutionally isomorphic to the diagnosis of "algorithmic xiangyuan" in Chapter 10: not only AI but also human regulators can be trained into structural blind spots within a similar incentive environment.

### §3.1.6 Falsifiability Condition

Diagnosis I can be falsified if one can find a financial case in which $\tau_g/\tau_c$ remained within the normal accumulation window for more than twenty years while the manifestation function $M(\tau)$ remained a linear climb and never underwent a nonlinear phase transition. Candidate cases for testing include:

- **Postwar Japan, 1945–1973**: the techno-capital clock $\tau_c$ of credit expansion was measured in years, while the governance clock $\tau_g$, administrative guidance by the Ministry of Finance, was measured in months to years, based on Hoshi & Kashyap (2010), giving $\tau_g/\tau_c \approx$ 1-10, $10^0$-$10^1$—a range that partially overlaps the normal accumulation window (roughly 3–10) and therefore cannot be excluded wholesale; since this chapter has not determined the business-cycle and governance-response starting points of that period under a unified caliber, the case at present constitutes neither a counterexample nor support and is listed as pending.
- **The United States, 1934–1980**, the Glass-Steagall era: the techno-capital clock $\tau_c$ was measured in years, credit expansion being suppressed by the Regulation Q ceiling on deposit rates (FDIC, 1998), and $\tau_g$ was also measured in years, the normal legislative cycle. Schularick & Taylor (2012) provide the long-run data. The ratio $\tau_g/\tau_c$ was about 1–3. This does not satisfy the condition of the normal accumulation window. A marked feature of this period is that the manifestation function $M(\tau)$ remained linear and large phase-transition events were absent, but that stability does not falsify Diagnosis I because the condition of the normal accumulation window was never satisfied.

This reverse test supports one implicit conclusion of Diagnosis I: **the very existence of Scenario I is itself a product of post-1980s financial liberalization**. The reason the U.S. financial system in the Glass-Steagall era did not experience a 2008-style phase transition is not that regulation was "better," but that the structure of the clock ratio $\tau_g/\tau_c$ itself did not fall into Scenario I. The speed of the projection side and the speed of the defense side remained at the same order of magnitude. This observation reformulates the standard "speed thesis," that regulation must catch up with innovation and allow some safe lead time, into a "structural thesis": once $\tau_g/\tau_c$ shifts from $10^0$ to the order of $10^{0.5}$-$10^1$, an accumulation window is automatically established, and there is no such thing as a "safe lead time." The essence of financial liberalization is not that it lets innovation run faster than regulation, but that it accelerates $\tau_c$ without paired acceleration of $\tau_g$, thereby pushing the system from $\tau_g/\tau_c \approx 10^0$ into Scenario I.

The incremental value of Diagnosis I exists only in contemporary finance, the era in which Scenario I is established. This is a counterexample to the methodological rule that "if the conclusion still holds after removing the MCAF variables, it is not written": once the structural distinction between the techno-capital clock $\tau_c$ and the governance clock $\tau_g$ is removed, this diagnosis collapses into an ex post comment that "regulators reacted too

slowly in 2008," losing predictive significance.

**Relation to the existing literature.** The core phenomenon of Diagnosis I, the sudden explicitation of implicit risk in September 2008, has already been described with precision in the existing literature on financial crises. Gorton's (2010) concept of "information sensitivity" points out that once market participants' uncertainty about asset quality exceeds a certain threshold, the asset jumps from "information-insensitive," able to circulate as collateral, to "information-sensitive," which nobody is willing to hold, thereby triggering liquidity collapse. Brunnermeier's (2009) model of the "liquidity spiral" describes the same dynamics: decline in asset prices -> margin calls -> forced sales -> further decline in prices, a self-reinforcing loop.

The relation of MCAF to these existing models is one of **formal complementarity rather than substantive replacement**: Gorton's "information-sensitivity jump" corresponds in MCAF to $M(\tau)$ moving from its gentle segment into its steep segment; Brunnermeier's "liquidity spiral" corresponds to the positive-feedback amplification mechanism of the slope of $M(\tau)$, manifestation itself triggering more manifestation.

MCAF's increment lies in linking these dynamic mechanisms causally to **the accumulation window of the clock ratio** $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$, and in pointing out that the critical phase transition is not an isolated feature of 2008, but the inherent structural consequence of Scenario I, the normal accumulation window. Any financial system that satisfies $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$ for a sufficiently long duration will face a similar risk of phase transition in $M(\tau)$. Existing literature is already highly complete at the level of mechanism in analyzing the 2008 crisis; MCAF's contribution is to reorganize these mechanisms into a **structural framework for cross-case comparison**, making it possible to understand the United States in 2008, LTCM in 1998, and future systemic risk in AI within the same coordinate system.

---

### §3.2 Diagnosis II: The Impossibility of Pure Defense-Side Intervention (Primary Evidence: Japan in the 1990s)

**Diagnostic proposition**: once the Cognitive Reynolds Number $Re_c$ has entered the high-risk interval and shadow debt $D_c^{shadow}$ has solidified within unsealed inventory $C$, an intervention strategy that merely increases the defense side, meta-audit $A$ and legitimacy base $\|L\|$, while leaving the projection side, $C$ and media conductivity coefficient $\Lambda$, untouched, is **structurally incapable of liquidating already deposited shadow debt**. Such a strategy can only slow the manifestation speed of the manifestation function $M(\tau)$; it cannot change the long-tail distribution of the post-manifestation settlement. Japan's policy path from the 1990s until before 2002—capital replenishment, deposit-insurance strengthening, and monetary easing systematically deployed without touching problem-asset stock—constitutes the sample for this diagnosis (after the 2002 Takenaka Plan opened stock disposal, the nature of the path changed; see §3.2.4): bank recapitalization (on the loss-absorption side; see the classification discussion in §3.2.4), strengthening of deposit insurance on the procedural side of $\|L\|$, and monetary easing as a buffer on the side of performance legitimacy $L_{perf}$ were all systematically deployed, but the "zombie loans" already sedimented in bank balance sheets, directly corresponding to the $C$ side, were left untouched. The result was a post-manifestation long-tail payout of roughly 14–25 years.

#### §3.2.1 Background Facts

| Phase | Time | Key event | MCAF category |
|---|---|---|---|
| Bubble formation | 1985–1989 | Post-Plaza Accord easing -> land/Nikkei doubling | Rapid generation of unsealed inventory $C$ |
| Bubble burst | 1990–1991 | Rate hikes + quantity controls -> Nikkei -49%, land declines | Manifestation function $M(\tau)$ starts |
| Regulatory forbearance | 1991–1997 | NPL accumulation, jusen crisis (¥685 billion public intervention) | Pure defense side, meta-audit $A$ inactive |
| Institutional rebuilding | 1998–2003 | Financial Reconstruction Act, zero interest rate, Takenaka Plan, Resona nationalization | Defense side, proceduralization of legitimacy base $\|L\|$ |
| Repair concluded | 2005 | Major-bank NPL ratio falls to 2.9% | Governance-clock $\tau_g$ cycle completed |
| Long-tail payout | 2005–2010s | TFP loss persists, fiscal deficit/GDP > 5% | Post-manifestation settlement still running |

(Data sources: Hoshi & Kashyap, 2010; Koo, 2008; Caballero et al., 2008; Ito & Mishkin, 2004)

### §3.2.2 The Triple Function of the Ministry of Finance and the Coupling Failure of Meta-Audit $A$ at the Level of the Legitimacy Base $\|L\|$

Before the 1980s, Japan's Ministry of Finance, MOF, bore a triple function: financial regulation, meso-audit $A_{meso}$; fiscal-policy making, direct management of performance legitimacy $L_{perf}$; and maintenance of ideological legitimacy, the institutional anchor of $L_{ideo}$, the reputational capital of "the MOF never makes mistakes." The coupling of these three functions was the basis of the stability of Japan's financial system in the 1960s-1980s. The authority of the MOF as regulator rested on the success of its fiscal policy and on its ideological legitimacy, while financial institutions' obedience to the MOF rested on this whole legitimacy package.

After the bubble burst of the 1990s, however, this coupling became **an amplifier of regulatory failure** rather than a stabilizing mechanism. In crisis, the MOF's three functions cancelled one another out. As meso-audit regulator $A_{meso}$, it needed to expose the banking system's non-performing loans so as to push liquidation forward; as fiscal-policy maker responsible for performance legitimacy $L_{perf}$, it needed to avoid exposing non-performing loans so as to maintain market confidence and prevent further decline in $L_{perf}$; as the ideological anchor of $L_{ideo}$, it needed to maintain the narrative of "the superiority of Japan's financial system" so as to preserve its own reputational capital. The mutual cancellation of these three functions after the bubble is the **structural reason** for the MOF's sluggish policy response in the 1990s, not a matter of individual decision error.

The MCAF significance of this observation is that the coupling of meta-audit $A$ and legitimacy base $\|L\|$ within a single institution may provide stability during normal periods, but becomes a structural vulnerability during crisis. When one single institution simultaneously bears meta-audit $A$ and multiple dimensions of $\|L\|$, perfor-

mance and ideology, a challenge to any one dimension threatens the legitimacy of the institution as a whole, thereby inducing delay in exposing the true state of affairs. This structural diagnosis has direct implications for contemporary regulatory architecture: the separation of the Federal Reserve and the SEC in the United States, the 2013 split of the United Kingdom's Financial Services Authority (FSA) into the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA), and China's former "one bank and three commissions" (merged into "one bank and two commissions" in 2018, and after the 2023 reform reconstituted into the plural structure of the Central Financial Commission, the National Financial Regulatory Administration, and the CSRC) can all be understood as institutional arrangements intended to avoid precisely this coupling failure.

#### §3.2.3 Zombie Lending and the Inertia on the Side of Unsealed Inventory $C$

Caballero et al. (2008) define zombie firms as firms that survive only because banks continuously roll over their loans and extend new credit to pay off the old, though they would be unable to cover even interest payments if financed at market rates. Their empirical estimates show that from the mid-1990s to the early 2000s the share of zombie firms among listed Japanese companies lay between 15% and 30%, especially concentrated in construction, real estate, and retail. Peek & Rosengren (2005) add, from the angle of bank incentives, that Japanese banks' continued lending to zombie firms was not a passive consequence of regulatory oversight failure, but an active choice under the incentive to limit the growth in reported problem loans so as to maintain required capital ratios. This incentive structure gave misallocation of resources institutional persistence.

Within the MCAF framework, zombie loans are the **institutionalized manifestation of already deposited shadow debt** within unsealed inventory $C$. On the books, banks label these loans "performing," but their true probability of default is far higher than the balance sheet suggests. Shadow debt is hidden under the outer shell of "normal loans." This concealment cannot be reduced to individual fraud by banks—the alignment of incentives among the MOF, the banks, and the borrowing firms under their shared interest in "avoiding manifestation" made deferred recognition a persistent institutional state (a structural explanation does not exclude individual violations): the MOF does not wish to see bank capital adequacy ratios fall massively below standard, threatening performance legitimacy $L_{perf}$ and ideological legitimacy $L_{ideo}$; banks do not wish to consume their own capital by writing down loans, threatening their continued existence; and borrowing firms do not wish to be declared bankrupt. The alignment of interest among these three parties gives zombie loans institutional continuity.

In MCAF terms, this hiding mechanism belongs to **administrative concealment**: regulators choose not to look; the information is technically accessible, since the true condition of bank balance sheets can be obtained through strict audit, but the exercise of audit power is suppressed by the multidimensional constraints of the legitimacy base. By contrast, the concealment of CDO risk in the United States in 2008 belongs to **structural encapsulation**: nesting and model dependence markedly raised the cost of identification—the correlation structure of underlying assets was doubly obscured by mathematical models and legal layers of insulation, and regulators could hardly penetrate the multilayer nesting structure of CDO$^2$ within the existing constraints of time and information. The common structure of the two concealment mechanisms is that both keep $D_c^{shadow}$ institutionally unobservable; the difference lies in the level at which blockage occurs. In administrative concealment the blockage lies in the willingness of $A_{meso}$ to exercise power, whereas in structural encapsulation it lies in the ability of $A_{meso}$ to exercise power. This distinction is independently testable beyond Japan and the United States. The AI scenario may contain a third variant, "technically accessible but cognitively inaccessible," as under a

model-filing regime where code is open yet complexity exceeds the auditor's $E_c$ ceiling. That possibility is left to later discussion in Chapter 15.

The key finding of Caballero and others is that the existence of zombie firms is not merely passive "uncleared loss," but an **active crowding-out** of healthy firms. Surviving on banks' continued rollover, zombie firms block the exit of capacity and factors of production: their profit-insensitive low-price sales depress product prices across the entire industry, while their continued occupation of labor and other factors sustains factor demand—the two together compress the profit space of healthy firms and suppress their entry, investment, and expansion. This crowding-out mechanism means that Japan's economic stagnation in the 1990s was not only a matter of "existing problems left unresolved," but also of "future growth being suppressed." The solidification of shadow debt within unsealed inventory $C$ generates damage to the economy that is **cumulatively ongoing**, not a one-shot shock.

### §3.2.4 The Three Stages of Pure Defense-Side Intervention and Their Failure

The interventions adopted by the Japanese government between the 1990s and 2005 can be classified by MCAF variables:

- **Intervention on the loss-absorption side**: recapitalization of banks in the 1990s—raising loss-absorption capacity, which under MCAF should be recorded as support for the redemption capacity of unsealed inventory ($C$) rather than as meta-audit ($A$)—with about ¥1.8 trillion injected into 21 major banks in March 1998 under the Financial Function Stabilization Act and about ¥7.5 trillion into 15 banks in March 1999 under the Financial Function Early Strengthening Act of October 1998; together with strengthened accounting-disclosure requirements (including the introduction in 1999 of internationalized impairment standards) and BIS capital-adequacy supervision (the latter two belonging to the meta-audit side).
- **Procedural devices on the side of legitimacy base** $\|L\|$: expansion of the authority of deposit-insurance institutions in 1996; establishment of the Financial Reconstruction Commission in 1998; and establishment of the Resolution and Collection Corporation in 1999.
- **Buffering on the side of performance legitimacy** $L_{perf}$: the zero-interest-rate policy beginning in 1999, QE beginning in 2001, and fiscal stimulus, with multiple rounds of economic packages over some fifteen years from the 1990s onward cumulatively totaling about ¥140 trillion (Koo, 2008).

These interventions stabilized the surface order of the financial system between 1998 and 2002. There was no large-scale bank run; the full guarantee arrangement for bank deposits was maintained, and the client assets and trade settlement of Yamaichi Securities, which collapsed in 1997, were also handled in an orderly manner; systemic collapse was avoided. But these interventions **did not touch the side of unsealed inventory** $C$. Zombie loans continued to be rolled over, and the ratio of non-performing loans on bank balance sheets kept rising from the late 1990s into the early 2000s, rather than falling — the ratio of major banks' non-performing claims disclosed by Japan's Financial Services Agency rose from about 6% at the end of fiscal 1998 to a peak of 8.4% at the end of fiscal 2001 (FSA, 2009). A caveat is needed: a rising NPL ratio may reflect both new losses and tightened recognition standards; judging whether disposal was effective requires combining new NPL formation, write-offs and transfers, and changes in classification standards, not the rise or fall of the disclosed ratio alone.

The turning point was the Takenaka Plan of October 2002. Building on the nationalization of individual banks in 1998, it for the first time set the identification and reduction of major banks' problem assets as an industry-wide hard target with a timetable—a systematic forced liquidation of unsealed inventory $C$. The plan required major banks to identify non-performing loans according to strict self-assessment standards, halve their non-performing-loan ratios within two years, and allowed the government to inject capital directly in exchange for bank equity. The nationalization of Resona Bank in 2003 became the emblematic event of this plan: the government effectively took over a major bank and forced it to value non-performing loans according to true value.

The reason the Takenaka Plan could be implemented in 2002–2005 whereas similar intervention could not be implemented during the 1990s itself reflects change in the legitimacy base $\|L\|$. After more than ten years of stagnation, the persuasiveness of performance legitimacy $L_{perf}$, the narrative of economic growth, had been weakened by prolonged stagnation; ideological legitimacy $L_{ideo}$, the superiority of Japan's financial system, had already been falsified by facts; and the coupling of the MOF's triple function had already disintegrated, since in 1998 its bank-regulatory function was stripped off and transferred to the independent Financial Supervisory Agency. This change in $\|L\|$ should be explained by the joint action of economic pressure, organizational reform, and policy choice—the adjustment of regulatory functions and disposal policies changed the conditions of action at the same time, and it should not be attributed to a natural erosion of legitimacy.

### §3.2.5 The Long-Tail Payout After Manifestation

The post-manifestation settlement horizon of Japan's Lost Decade was not ten years but 14–25 years. Financial-system repair was completed in stages in 2005, with the non-performing-loan ratio falling to about 2.9%, but this was only the liquidation of observable debt $D_c^{obs}$. Other dimensions of true debt $D_c^{true}$ continued to pay out:

- **TFP losses**: Hayashi & Prescott (2002) show that Japan's TFP growth dropped abruptly from an annual average of 2.4% in 1983–1991 to an annual average of 0.2% in 1991–2000 (the data of that paper end in 2000; this chapter does not cite it for judgments about the 2020s).
- **Changes in labor structure**: the share of non-regular employees among Japanese male workers aged 35–54 rose from 3.7% in 1992 to 9.8% in 2012, with the share of regular employment correspondingly falling from 96.3% to 90.2% (Statistics Bureau of Japan, Employment Status Survey). The long-term consequences of expanding non-regular employment include lower household-formation rates, lower fertility, and inadequate pension preparation.
- **Accumulation of fiscal liabilities**: Japan's government-debt-to-GDP ratio rose from about 63% in 1990 to about 175% in 2005, and reached about 248% in 2022 (IMF *World Economic Outlook* database). The sustained expansion of public finance ran in parallel with the liquidation of private-sector losses, suggesting a component of loss transfer into the public sector—the scale and mechanism of the transfer remain a hypothesis to be tested, and this chapter does not equate a rising debt ratio with loss transfer.
- **Demographic feedback**: the decline of household formation and fertility in the 1990s entered the working-age population through a twenty-year cohort effect (the contraction of whose total size had already begun in the late 1990s), further suppressing TFP growth. Twenty years later the long-run consequences of shadow debt feed back into the basic conditions of the real economy through demographic structure itself.

The cumulative payout across these dimensions exceeds the control range of any single financial policy. Once the post-manifestation settlement enters its long tail, it cannot be shortened merely by accelerating the governance clock $\tau_g$, not because repair is insufficiently thorough, but because the post-manifestation settlement and the governance response are not synchronized. Standard analysis uses "the clearing of non-performing loans" as the sign that crisis is over; MCAF identifies this sign as an analytical illusion. It marks the completion of the $\tau_g$ cycle, not the completion of the post-manifestation settlement. Rapid policy launch can stop losses from continuing to expand, but it does not mean that the damage already formed disappears in step; whether intervention shortens the repair horizon depends on its timing, object, and manner of execution (the cross-case comparison in §3.2.6 is based precisely on this). This is the supplement of Diagnosis II to Diagnosis I: Diagnosis I shows that during the phase-transition period of the manifestation function $M(\tau)$, $\tau_g$ is compressible; Diagnosis II shows that during the long-tail period of the post-manifestation settlement, compressing $\tau_g$ alone cannot resolve the matter. The two describe different degrees of manipulability in the two phases of pre-manifestation waiting and post-manifestation settlement.

### §3.2.6 Falsifiability Condition

Diagnosis II can be falsified if one can find a case in which the Cognitive Reynolds Number $Re_c$ has entered the high-risk interval and shadow debt $D_c^{shadow}$ has already solidified within unsealed inventory $C$, yet the debt is successfully liquidated **purely through defense-side intervention**, without touching $C$. Candidate cases include:

- **The treatment of the U.S. savings-and-loan crisis after the 1987 crash, 1989–1995**: the Financial Institutions Reform, Recovery, and Enforcement Act, FIRREA, established the Resolution Trust Corporation (RTC) in 1989, which took over all 747 failed savings institutions, disposed of assets with a book value exceeding $458 billion, and recovered about $397 billion for taxpayers (RTC, 1995). This was a direct liquidation of unsealed inventory $C$, not a pure defense-side intervention, and therefore does not falsify Diagnosis II.
- **The treatment of the Swedish banking crisis, 1991–1993**: the Swedish government nationalized its two largest banks, Nordbanken and Götabanken, and established bad banks to strip non-performing assets. This was a direct liquidation of $C$ and does not falsify Diagnosis II.
- **The treatment of the Korean financial crisis, 1997–1999**: the Korean government forcibly implemented large-scale asset restructuring of major banks and debt haircuts for chaebol groups, deploying about KRW 168.6 trillion, around $135 billion, of public funds to dispose of non-performing assets (Chopra et al., 2002; Financial Services Commission of Korea, 2009). This too was liquidation of $C$ and does not falsify Diagnosis II.

These three historical cases share one common feature: **all of them touched unsealed inventory** $C$. Either they imposed forced bankruptcy liquidation or forced asset stripping. Their post-manifestation settlement horizons were significantly shorter than Japan's, about 5–7 years in the United States, 3–4 years in Sweden, and 3–5 years in Korea (Chopra et al., 2002; Englund, 1999; RTC, 1995), supporting the reverse inference of Diagnosis II: **intervention that touches** $C$ **can shorten the post-manifestation settlement horizon**.

Diagnosis II therefore has a clear control group in financial history: all known cases of "pure defense-side intervention," leaving unsealed inventory $C$ untouched, are accompanied by long post-manifestation settlements; all known cases of "liquidative intervention," touching $C$, are accompanied by short post-manifestation settle-

ments. This contrast is not a one-to-one monocausal proof, but it strongly reinforces the structural assertion of Diagnosis II.

**Relation to the existing literature.** The core conclusion of Diagnosis II, that bank recapitalization without clearing bad assets cannot shorten the post-manifestation settlement horizon, has been repeatedly confirmed in the literature on Japan's Lost Decade. Caballero, Hoshi & Kashyap's (2008) zombie-lending model shows precisely that banks' continued rollover to zombie firms not only prolonged repair of bank balance sheets themselves, but also suppressed economy-wide reallocation of resources through the crowding-out effect. Hoshi & Kashyap (2010) characterize Japan's policy response overall as regulatory forbearance and distill from it eight lessons—the first being that banks may refuse public funds (the instance of Japan's 1998 recapitalization program), the second that programs of asset purchase and recapitalization must be big enough; forbearance prolonged the crisis rather than mitigating it. Peek & Rosengren (2005) provide empirical evidence that continued lending support by Japanese banks to related firms significantly reduced productivity growth in the industries where those firms operated.

The relation of MCAF to these existing studies is one of **reorganization through variable mapping**: the distinction in Caballero and others between "zombie loans vs. asset clearing" corresponds in MCAF to "inertia on the inventory side of $C$ vs. forced liquidation of $C$ inventory"; Hoshi & Kashyap's proposition that forbearance prolongs crisis corresponds in MCAF to "intervening only on the defense side, $A/\|L\|$, while leaving the projection side, $C$, untouched cannot change the post-manifestation settlement horizon."

MCAF's increment is to elevate the distinction between "capital injection vs. asset clearing" from the level of policy tools to a **structural classification of projection side vs. defense side**, and, through comparison with the United States and Korea, where touching $C$ inventory produced post-manifestation settlements of roughly 5–7 years and 3–5 years respectively, to establish the cross-case regularity that "touching $C$ inventory can shorten the post-manifestation settlement horizon." Existing literature is already very deep in its mechanism analysis of Japan's policy failure; the contribution of MCAF is to map these mechanisms into the six-variable framework, allowing the Japanese case to be structurally compared with others in the two-dimensional space of "type of intervention, defense side vs. projection side x post-manifestation settlement horizon." That comparison elevates Diagnosis II from "Japan's special lesson" to the "general impossibility of pure defense-side intervention."

---

### §3.3 Diagnosis III: Structural Intervention and the Progressive Governance Clock (Primary Evidence: China's New Asset Management Regulations, 2018–2026)

**Diagnostic proposition**: when intervention acts simultaneously on the defense side, meta-audit $A$ and legitimacy base $\|L\|$, and on the projection side, unsealed inventory $C$, media conductivity coefficient $\Lambda$, and narrative load $\phi$, and when governance is designed on a **sustained scale**, measured in years or even a decade, rather than a **pulsed scale**, measured in days to weeks, then the slope of the manifestation function $M(\tau)$ can be actively controlled so that the manifestation and liquidation of shadow debt are completed without entering a critical phase-transition state. The path of China's New Asset Management Regulations from 2018 to 2026 is the primary evidence for this diagnosis: one structural intervention directed at about RMB 100 trillion of shadow-banking inventory, implemented through a governance persistence period lasting nine years (2017–2026, still deepening as of 2026), has enabled large-scale transfer of shadow debt $D_c^{shadow}$ into observable

debt $D_c^{obs}$ without a systemic financial crisis—whether liquidation has been completed requires distinguishing among loss recognition, risk transfer, and deferred recognition (the falsifiability condition of §3.3.6 targets precisely this). This path is not the result of "slow regulation," but of the **active design of the governance persistence period** (what was set in advance was the transition arrangement; the 2020 extension was a recalibration in response to the pandemic). The core increment of Diagnosis III is to reinterpret governance from a one-off response into an **institutional variable that can itself be designed** (the governance clock $\tau_g$ retains the response-cycle meaning of Chapter 3 §3.2; the duration of sustained implementation after the response is distinguished in this chapter by the descriptive name "governance persistence period").

The difference between this diagnosis and the existing Chinese literature on financial regulation is the following. Existing analyses, for example the public speeches of central-bank leaders such as Yi Gang (2018) and Guo Shuqing (2018), use a narrative framework of "preventing and defusing major financial risks" plus a list of concrete policy instruments. This is a **policy narrative** form. MCAF's increment is to reorganize the New Asset Management Regulations from one concrete policy into an intervention by the design of the governance persistence period upon the slope of the manifestation function $M(\tau)$, and to place it in a triangular contrast with §3.1, the pulsed $\tau_g$ of the United States, and §3.2, the pure-defense $\tau_g$ of Japan, thereby forming a **trinitarian typology of financial governance clocks**.

### §3.3.1 The Generative Mechanism of Shadow Banking and the Inventory Scale in 2017

The core form of China's shadow-banking system is not a U.S.-style securitization chain of mortgage-backed securities (MBS) and collateralized debt obligations (CDOs), but a multilayer nesting structure **with bank wealth-management as the entry point, trust plans / broker asset management / fund-subsidiary channels as the transmission path, and non-standardized credit claims as the asset end**. The historical starting point of this structure lies around 2010. After the large-scale credit expansion produced by the 2008 "four-trillion" stimulus, loan-scale controls began to constrain the system in 2010, and banks responded by shifting on-balance-sheet loans into off-balance-sheet wealth management so as to evade credit-quota management, thereby generating the first generation of shadow-banking products.

The time series of scale data is as follows. Outstanding bank wealth-management products in China stood at about RMB 2.8 trillion at the end of 2010, about RMB 10 trillion at the end of 2013, about RMB 23.5 trillion at the end of 2015, and reached about RMB 29 trillion at the end of 2016. During the same period, trust assets rose from about RMB 3 trillion at the end of 2010 to about RMB 20 trillion at the end of 2016; broker asset management rose from about RMB 2 trillion at the end of 2012 to about RMB 17.6 trillion at the end of 2016; and fund-subsidiary special accounts rose from zero to about RMB 10.5 trillion at the end of 2016.

If these transmission segments are added together and double counting is deducted, Moody's (2018) estimates total Chinese shadow banking at the end of 2017 at about RMB 65 trillion, about 79% of GDP. The public "broad shadow banking" caliber of the China Banking and Insurance Regulatory Commission, first released in 2020, estimates that broad shadow banking reached a historical peak of about RMB 100 trillion at the beginning of 2017, with its ratio to GDP falling from 123% at the end of 2016 to 86% at the end of 2019. The gap between the two calibers arises mainly because Moody's deducts the double counting among nested SPVs, whereas the broad caliber of the banking-and-insurance regulator includes the entire stock of banks' off-balance-sheet commitments and non-standardized credit claims.

Under either caliber, the scale of Chinese shadow banking relative to GDP around 2017 had already entered the high-risk interval. This is the quantitative criterion within MCAF for the accumulation of $C \cdot \Lambda$. As a comparison, the gross measure of U.S. shadow-banking liabilities peaked at nearly $22 trillion in June 2007, exceeding the roughly $14 trillion of total traditional-banking liabilities at the time (Pozsar et al., 2013).

**The structure of the mapping chain.** A typical shadow-banking mapping chain includes 3–5 layers of nesting. Take the funds raised by a bank wealth-management product as an example. At the first layer, these funds subscribe to a targeted asset-management plan of securities company A, SPV A. At the second layer, SPV A subscribes to the single-fund trust of trust company B, SPV B. At the third layer, SPV B contributes capital to an asset-management plan of fund-subsidiary company C, SPV C. At the fourth layer, SPV C extends an entrusted loan to, or purchases a private bond issued by, the local-government financing platform. The function of these four layers of nesting is to evade bank credit-concentration limits on local-government financing platforms and evade lending restrictions on specific industries such as real estate or sectors with overcapacity—where the underlying debtor is not sufficiently identified through penetration, institutions exploit product-classification differences to evade credit and investment limits; to reduce capital occupation (different accounting and regulatory classifications correspond to different capital-charge requirements, not a blanket exemption); and to aggravate maturity mismatch (each SPV can be designed as a short-term product matched against long-term assets).

**Narrative load $\phi$.** The core narrative that maintains the operation of this mapping chain is "implicit guarantees." Even where wealth-management product contracts explicitly state "non-principal-protected floating yield," the shared expectation of market and investors is that banks will ensure full repayment of principal and contractual return when products mature. The narrative of implicit guarantees is not a legal clause, but a **narrative equilibrium** consolidated jointly by uniform market expectation and the actual conduct of issuing banks across multiple events. Between 2013 and 2017 a number of trust products experienced risks of default, and many were prevented from immediate default through backstopping by banks or trust companies using their own funds, or through extensions and rollovers—this chapter cites publicly reported cases among them as process evidence and does not attempt an exhaustive enumeration. Along the dimension of $\phi$, the implicit-guarantee narrative gave wealth-management products a perceived risk profile close to bank deposits, while their returns, annualized at 4%–5%, significantly above the roughly 2.75% rate on three-year time deposits, maintained continuing attractiveness relative to deposits.

**Regulatory failure on the dimension of meta-audit $A$.** Before 2017, China's financial-regulatory architecture adopted the separated model of "one bank and three commissions," the central bank, the banking regulator, the securities regulator, and the insurance regulator, with each regulatory body supervising only the institutions within its own jurisdiction. The cross-institutional nesting structure of shadow banking exploited precisely the fissures of this architecture. Banking-side regulation, by the CBRC, saw only the issuer of the wealth-management product; trust-side regulation, by the trust-business department of the banking regulator, saw only the manager of the single-fund trust; securities-side regulation, by the CSRC, saw only the asset-management plan. Information on the complete chain was dispersed across different institutions and departments, and the difficulty of look-through identification and responsibility coordination rose systematically. This regulatory failure was not a matter of "insufficient regulatory capability," but a **structural defect in the regulatory architecture itself**. The incompleteness of meta-audit $A$ was the institutional precondition of the expansion of shadow banking, not its passive outcome.

### §3.3.2 The Institutional Design of the New Asset Management Regulations: Synchronized Intervention on $\Lambda$, $\phi$, and $A$

On November 17, 2017, the People's Bank of China, the China Banking Regulatory Commission, the China Securities Regulatory Commission, the China Insurance Regulatory Commission, and the State Administration of Foreign Exchange ("one bank, three commissions, and one administration") jointly released the Guiding Opinions on Regulating the Asset Management Business of Financial Institutions (Exposure Draft). In March 2018, the State Council's institutional reform merged the banking regulator and the insurance regulator into the China Banking and Insurance Regulatory Commission, so that when the formal version was released on April 27, 2018, the issuing bodies became "one bank, two commissions, and one administration"; this is the "New Asset Management Regulations" (People's Bank of China et al., 2018). In July 2020 the transition period was extended from the originally planned end of 2020 to the end of 2021, and on January 1, 2022, the unified transition period ended and the New Asset Management Regulations entered the stage of full enforcement.

The core clauses of the New Asset Management Regulations can be classified by dimension of intervention under the MCAF mapping:

**Intervention on the media conductivity coefficient $\Lambda$, chain length.** Item 22 restricts channel business: asset-management products may reinvest in one additional layer of asset-management product, but the invested product may not itself reinvest in asset-management products other than public securities investment funds. This clause converges the nesting as follows: an asset-management product may reinvest in at most one additional asset-management product, and the invested product may only reinvest in public securities investment funds—whether the starting product is counted must be specified, and the rule cannot be summarized as "one public fund plus one private product" (compressed from the original 3–5 layers to at most 2). Item 22 simultaneously prohibits financial institutions from providing channel services to other institutions' asset-management products for the purpose of evading investment-scope restrictions or leverage constraints. This intervention is a direct compression of the $\Lambda$ dimension. It does not lower $\Lambda$ by "raising standards," but by **physically limiting the upper bound of** $\Lambda$ so that it returns to a penetrable interval.

**Intervention on narrative load $\phi$.** Item 2 explicitly breaks implicit guarantees: financial institutions engaged in asset-management business may not promise principal preservation or guaranteed returns; where redemption difficulties occur, financial institutions may not advance funds for redemption in any form. Item 19 defines the determination scenarios of implicit guarantees and sets two categories of sanctions: where a deposit-taking financial institution engages in implicit guarantees, it is deemed to have carried out regulatory arbitrage by means of an asset-management product with the essential character of a deposit, and must pay in full the deposit reserves and deposit-insurance premiums owed, and is subject to administrative punishment; where a licensed non-deposit-taking financial institution engages in implicit guarantees, it is deemed to be operating in violation, is corrected according to law, and is punished. Item 18 complements this with net-value management: asset-management products must implement net-value management; the generation of net value must conform to corporate accounting standards, timely reflecting the return and risk of the underlying financial assets. This set of interventions is not a soft constraint of "advising investors to be prudent." Rather, by **prohibiting financial institutions from performing implicit-guarantee behavior in any form**, it eliminates from the supply side the institutional carrier of the implicit-guarantee narrative. The decline of $\phi$ is not achieved by persuading investors to change expectations, but by making banks cease to provide implicit guarantees in practice, so that implicit-guarantee expectations naturally lose force.

**Intervention on meta-audit $A$.** Item 15 prohibits fund-pool operation: each asset-management product must be separately managed, separately booked, and separately accounted for; rolling issuance, collective operation, and separated pricing characteristic of fund-pool business may not be carried out or participated in. This clause eliminates one of the most common transparency defects of the earlier regime, namely the adjustment of funds among different products within the same wealth-management pool. Under fund-pool operation, the true balance sheet of a single product cannot be penetrated, because the product can at any time borrow liquidity from other products in the same pool. Once fund pools are prohibited, separate management, booking, and accounting improve the traceability of each product's asset attribution and fund flows, providing a necessary condition for look-through supervision—this does not automatically guarantee that underlying risks are fully visible. Item 25 establishes a unified reporting system for asset-management products, under which the People's Bank of China takes charge of the data coding and consolidated statistics of asset-management products: basic and initial fundraising information is to be reported within 5 working days of the product's establishment, fundraising and asset-and-liability information during the continuation period before the 10th of each month, and termination information within 5 working days of the product's termination.

**The structure of the timeline.** The design of the timeline of the New Asset Management Regulations is itself an embodiment of the governance persistence period:

- November 2017: exposure draft released, the governance clock $\tau_g$ enters its startup phase.
- April 2018: formal version released, with a transition period from April 2018 through December 2020, about 32 months.
- April 2018-July 2020: first-stage transition period, during which financial institutions begin gradually compressing legacy products and issuing net-value products that comply with the new rules. The scale of wealth-management products, from about RMB 29 trillion at the end of 2016, did not immediately decline significantly, but the product structure underwent qualitative change: the share of net-value products rose from about 14% before the release of the New Asset Management Regulations (on the non-principal-guaranteed caliber, derived from the depository center's multi-year data) to about 55% in June 2020 (China Wealth Management Registration and Depository Center, 2021, 2022).
- July 2020: because of the impact of COVID-19, the transition period is extended to December 2021, an extension of one year. This extension is not a compromise on regulatory goals, but an **active recalibration** of the governance persistence period: when an exogenous shock disturbs the pace of stock disposal, the persistence period is correspondingly lengthened so as to avoid triggering a critical phase transition of the manifestation function $M(\tau)$.
- January 2022: the unified transition period ends and the New Asset Management Regulations enter full enforcement. By the end of 2021, the share of net-value products had reached about 93%; investment in non-standard assets declined continuously, with the share of non-standardized credit claims held by bank wealth-management products falling 14.8 percentage points from the time the New Asset Management Regulations were issued (to the end of 2021) (China Wealth Management Registration and Depository Center, 2022); and the scale of broad shadow banking contracted from its historical peak of about RMB 100 trillion at the beginning of 2017 to about RMB 84.8 trillion at the end of 2019 (Research Group of the Policy Research Bureau of the CBIRC, 2020).

The structural feature of this timeline is the **absence of any pulsed node**. There is no node like the U.S. case in September 2008, "crisis -> emergency legislation -> passage within days." The entire process from exposure

draft to full implementation lasted about 50 months, from November 2017 to January 2022, placing the governance persistence period on a scale measured in years, even up to five years. This scale stands in contrast to Diagnosis I in §3.1, where the U.S. governance clock after crisis was compressed to the scale of days. Both concern intervention upon shadow debt, yet the United States obtained a window of $\tau_g$ compression only after the critical phase transition of $M(\tau)$ had already occurred, whereas China, through the design of a sustained governance persistence period before criticality, prevented phase transition from ever being triggered.

**Process evidence for the death of the implicit-guarantee narrative.** The decline of narrative load $\phi$ was not accomplished instantaneously by policy announcement, but consolidated through the cumulative effect of multiple actual default events. Key stages of the process are these. A wave of corporate-bond defaults emerged in 2018–2019, 130 bonds defaulted in 2018 involving RMB 124.3 billion, about 219% higher than in 2017 (Financial Stability Analysis Group of the People's Bank of China, 2019). It must be noted that this default wave was not caused solely by the New Asset Management Regulations; deleveraging policy, China-U.S. trade frictions, and macroeconomic slowdown were also important drivers. The defaults are cited here only as process evidence of the solidified failure of the implicit-guarantee narrative, not for precise attribution of causal shares. In May 2019 Baoshang Bank was taken over, the first bank takeover in China in twenty years. In November 2020 Yongmei Holdings defaulted, and the value decline of Yongmei bonds held by multiple wealth-management products was reflected directly in those products' net values. In March and November 2022 two rounds of bond-market adjustment caused large-scale net-value breakage of bank wealth-management products, with net value falling below the initial benchmark of 1, and wealth-management companies did not compensate the broken products.

Zhang Yilin, Liang Wei, and Yu Yunjun (2023) argue from a micro-game-theoretic perspective that rigid payment, as an implicit rule within shadow banking, can only be broken by government-level coercive intervention. That theoretical prediction is consistent with the empirical process described above: before 2018, no matter how often policy stressed "breaking implicit guarantees," market expectations would not shift so long as individual institutions still backstopped at key points. Only after the New Asset Management Regulations institutionally prohibited such behavior across the whole system did $\phi$ materially decline.

This process on the dimension of $\phi$ reveals an important structural observation of MCAF: **investor education alone is insufficient to change principal-protection expectations; whether institutions still actually provide extraordinary redemption is the key condition for whether expectations can shift**. Before 2017, no matter how repeatedly regulators stressed "breaking implicit guarantees," the market-wide expectation of implicit guarantees would not disappear so long as some institutions continued to backstop at key moments. After 2018, only the institutional prohibition that caused implicit-guarantee behavior to systemically disappear across the financial system enabled market expectations truly to turn. This structural observation has direct analogical significance for discussions in AI regulation about "vendor self-regulation" and "narrative responsibility," which will be developed in §4.

### §3.3.3 The Persistence of Progressive Governance: Annual Evolution from 2022 to 2026

The end of the unified transition period for the New Asset Management Regulations in January 2022 is not the end of governance; it is the point at which governance moves from the "rule-establishment phase" into the "rule-deepening phase." The annual evolution from 2022 to 2026 allows the governance persistence period to be fully observed as an institutional feature.

**2022: transitional pain and the initial adaptation of investor behavior.** Chinese bank wealth-management products underwent two rounds of net-value breakage in 2022: in March, triggered by the adjustment of the equity market, and in November, triggered by the adjustment of the bond market (China Banking Association, 2023). These two rounds triggered waves of investor redemption: the total scale of wealth-management products across the market fell from a peak of about RMB 29.15 trillion in June 2022 to about RMB 27.65 trillion at year end (China Wealth Management Registration and Depository Center, 2023).

In MCAF terms, this phenomenon is the direct manifestation of adjustment on the dimension of narrative load $\phi$: for the first time, investors saw unpredictable swings directly in product net values and learned in practice what "non-principal-protected floating yield" means: no guarantee of principal. The important point is that **regulators made no compromise on the core clauses** even under the pressure of these two rounds of net-value breakage and large-scale redemptions. The principle of breaking implicit guarantees was not blurred again, net-value management was not postponed, and the prohibition on fund pools was not relaxed. This regulatory persistence under market pressure is the key structural feature of the governance persistence period: had regulators reintroduced any form of implicit-guarantee mechanism under the pressure of the 2022 breakages, the intervention on the dimension of $\phi$ within the whole architecture of the New Asset Management Regulations would have structurally collapsed.

**2023–2025: the deepening phase.** After the bond market stabilized in 2023, the scale of wealth-management products recovered to about RMB 26.8 trillion; by the end of 2024 it had risen back to about RMB 30 trillion, and by the end of 2025 it reached RMB 33.29 trillion, exceeding the June 2022 peak (China Wealth Management Registration and Depository Center, 2024, 2026)—showing that the scale retreat of 2022–2023 was mainly the result of cyclical redemption pressure rather than a permanent contraction of allocation. Scale changes must be analyzed together with funding sources, product structure, and valuation method, and cannot be used alone as an indicator of implicit-guarantee expectations or legitimacy change.

Regulation shifted from "prohibition" to "guidance," lengthening duration and increasing equity allocation, and the meaning of the governance persistence period expanded from "compressing the media conductivity coefficient $\Lambda$ / narrative load $\phi$" into "reconstructing the functional positioning of wealth-management products." In May 2023 the National Financial Regulatory Administration was established, replacing the former CBIRC and unifying supervision over banking, insurance, and nonbank financial institutions. In MCAF terms, this reform simultaneously raises meta-audit $A$ (look-through supervision of the financial sector excluding securities is now completed within a single institution, narrowing coordination gaps) and procedural legitimacy $L_{proc}$ (regulatory legal empowerment upgraded from the dispersed authorizations of "one bank and three commissions" to unified authorization), alleviating but not eradicating the "regulatory-arbitrage space" diagnosed by Zhong Zhen (2018)—securities-side regulation remains with the CSRC, so the arbitrage space along sectoral boundaries is narrowed rather than eliminated.

Through 2024–2025 regulation entered the phase of normalized deepening: the net-value product system stabilized; non-standard asset balances continued to contract; and the large-scale transfer of shadow debt $D_c^{shadow}$ into observable debt $D_c^{obs}$ advanced substantially (the distinction among risk reduction, transfer, and deferral is addressed in §3.3.6), while the governance persistence period continued.

**2026: unification of the information-disclosure regime.** The Measures for the Administration of Information Disclosure of Asset-Management Products of Banking and Insurance Institutions, promulgated on December 22,

2025 and effective from September 1, 2026, further institutionalizes the disclosure requirements for banking and insurance asset-management products. This node supports the judgment of "continued improvement of supporting institutions," but does not directly prove that shadow debt has been fully manifested or liquidated (National Financial Regulatory Administration, 2025).

**The institutional meaning of the governance persistence period.** The nine years from 2017 to 2026 are not the result of "ineffective enforcement." Compare the execution path of Dodd-Frank in the United States: legislation in 2010, major-rule making basically completed by 2015, about five years, and partial relaxation beginning in 2018 through the Economic Growth, Regulatory Relief, and Consumer Protection Act. By contrast, China's New Asset Management Regulations have operated for nine years with no relaxation of core regulatory standards, while supporting implementation arrangements were tested through the extension and the pandemic-period adjustments, and the regime has remained in a state of continuous deepening, manifesting a transition of governance from "one-off institution-building behavior" to a "persistent institutional state."

The MCAF meaning of this transition is that the governance persistence period is no longer merely the aftermath of "the regulatory response to a specific risk," but "the duration of sustained pressure maintained against a certain asset structure." Under the U.S. pulsed mode, the long gap after the passage of legislation provides conditions for the reaccumulation of shadow debt. Under the Chinese progressive mode, continuous pressure makes the reformation of shadow debt difficult.

### §3.3.4 The Coordinated Evolution of the Six MCAF Variables

Across the nine years of the New Asset Management Regulations, the six variables evolve as follows: the growth speed of unsealed inventory $C$ falls from fast to stable, indirectly constrained by channel restrictions; the media conductivity coefficient $\Lambda$ compresses from 3–5 nested layers to 2 or fewer, via Item 22; the narrative load $\phi$ is reconstructed from the shared expectation of "implicit guarantees and principal protection" to a cognition of "net-value fluctuation," through Items 18 and 19; meta-audit $A$ rises from the incapacity of separated supervision to penetrate, to unified registration plus look-through supervision, with Items 15 and 25 establishing product-level look-through capacity; the human biological substrate $E_c$ remains constant at 10 bits/s, the biological constraint; and the legitimacy base $\|L\|$ follows a U-shaped path, a "pseudo-high" in 2017 founded on implicit guarantees -> procedural legitimacy $L_{proc}$ raised by the 2018 institutional reforms—the Financial Stability and Development Committee, the merger into the CBIRC, and the transfer of macroprudential rule-making authority to the central bank—as recorded in *China Financial Stability Report 2019* -> impact on performance legitimacy $L_{perf}$ from the 2022 net-value breakages -> reconstruction in a new equilibrium by 2026.

Two observations deserve emphasis. First, the constancy of the human biological substrate $E_c$ imposes a real constraint on the New Asset Management Regulations: the success of net-value management in bank wealth management depended on simplified expression, a single net-value figure representing the current valuation of a product; net-value display lowers the threshold for recognizing current gains and losses, yet a single net-value figure cannot fully express liquidity, credit linkage, and tail risk—simplification of expression is not simplification of underlying risk. Second, the U-shaped movement of the legitimacy base $\|L\|$ reveals a structural tradeoff: the high legitimacy base in 2017 was a pseudo-high built upon implicit liabilities; the 2022 net-value breakages were an adjustment from pseudo-high down toward true level; the short-term cost in performance legitimacy $L_{perf}$ was the price paid for dissolving implicit liabilities; and by 2026 the credibility of a newly reconstructed legitimacy base is replacing the previous pseudo-high (the reconstruction segment has obtained initial support

on the scale dimension: the outstanding scale at the end of 2025, RMB 33.29 trillion, has already crossed the RMB 30 trillion threshold; the breakage-ratio dimension remains to be observed in subsequent market-stress periods—if the breakage ratio stabilizes below 5% under later stress, the reconstruction segment receives full support).

### §3.3.5 A Three-Dimensional Comparison with Japan and the United States

The following table integrates the structural comparison among the three diagnoses in §3.1, §3.2, and §3.3. It presents the structural differences among Japan in the 1990s, the United States in 2008, and China in 2018–2026 across the mode of the governance clock $\tau_g$, the form of the manifestation function $M(\tau)$, and the cost borne by the legitimacy base $\|L\|$.

| Dimension | Japan 1990s (Diagnosis II) | United States 2008 (Diagnosis I) | China 2018–2026 (Diagnosis III) |
|---|---|---|---|
| Timing of intervention | After bubble burst, after the critical point of $M(\tau)$ | After Lehman, in the middle of the critical phase of $M(\tau)$ | During accumulation of shadow debt, before the critical point of $M(\tau)$ |
| Scale of relevant risk-exposure vehicles (calibers differ; not directly comparable) | About 15%–25% of GDP (zombie loans + firms' own debt) | About $10 trillion scale (MBS/CDO/structured investment vehicles, SIVs) | About RMB 100 trillion (broad shadow banking) |
| Type of intervention | Pure defense side (without touching unsealed inventory $C$) | Emergency rescue + ex post legislation | Structural attack (simultaneously acting on $\Lambda/\phi/A$) |
| Feature of governance clock $\tau_g$ | Pulsed (concentrated in 1997–1999) + long stagnation period | Pulsed (concentrated in 2008–2010) + subsequent relaxation | Sustained (continuous deepening 2017–2026) |
| Slope of manifestation function $M(\tau)$ | Extremely slow (14-year slowly hidden manifestation) | Extremely steep (nonlinear jump within a few days) | Controllable (relatively even manifestation over an approximately 44-month transition) |
| Post-manifestation settlement horizon | 14–25 years (TFP loss still ongoing) | About 10–15 years (residual effect through the QE period) | Ongoing (expected 8–12 years) |
| Result | Lost Decade + demographic feedback | Political fragmentation + partial relaxation after 2018 | Progressive clearing, new equilibrium still forming |

(Table note: the three figures in the "scale of relevant risk-exposure vehicles" row are on mutually different calibers—Japan's GDP share is an estimate of non-performing claims, the U.S. figure is the notional balance of securitized products, and China's figure is the broad shadow-banking statistical caliber; all are risk-exposure-vehicle scales rather than direct measurements of shadow debt ($D_c^{shadow}$) balances, and per the layering discipline of Chapter 3 §7.6 they are not directly compared in quantity.)

**The core increment of Diagnosis III** is the structural difference revealed by placing the third column of this table beside the first two: under the sustained mode of governance, the slope of the manifestation function $M(\tau)$ can be actively controlled before a critical phase transition occurs, rather than only being passively responded to after criticality. This control capacity is not free. It requires that the governance persistence period be sustained on the scale of years or even a decade, that the regulatory architecture remain relatively independent of short-term market pressure over the long run, and that investors bear observable short-term losses during the period of adjustment in narrative load $\phi$. Whether these conditions exist in different institutional settings is open to discussion, but where they do exist, the structural possibility of this mode is established.

**Difference between existing analysis and MCAF analysis.** In the existing Chinese literature discussing the 2020 extension of the transition period (Xiao Gang, 2020), the mainstream interpretation treats it as a balancing arrangement between stable growth and risk prevention; earlier research on the financial cycle and orderly adjustment (Peng Wensheng, 2017) provides the mechanistic background for understanding phased governance. This is correct at the level of policy-effect description, but it implicitly treats the extension as a compromise of a "non-ideal state." The MCAF perspective offers a different reading: **extension is an endogenous feature of the design of the governance persistence period**. When the pace of stock disposal is disturbed by exogenous shock, the persistence period adjusts accordingly so that the manifestation function $M(\tau)$ does not trigger a critical phase transition. Only the structural reading offered by MCAF can explain why "after extension, the core clauses were not compromised", breaking implicit guarantees, net-value management, and the prohibition of fund pools were not relaxed. These clauses are the core of intervention on $\Lambda$, $\phi$, and $A$; relaxing them would cause the entire structure of the governance persistence period to collapse. The structural distinction between lengthening the timeline, negotiable, and maintaining the core clauses, non-negotiable, is the key increment Diagnosis III adds over the existing policy narrative.

### §3.3.6 Falsifiability Condition

Diagnosis III can be falsified if one can find a case in which governance is designed as sustained (with the persistence period measured in years or longer) and simultaneously intervenes upon the media conductivity coefficient $\Lambda$, narrative load $\phi$, and meta-audit $A$, yet the manifestation function $M(\tau)$ still undergoes a critical phase transition. If other paths, such as pulsed disposal, can also complete liquidation without triggering a critical phase transition, this shows that progressive governance is not a necessary condition—this narrows its scope of applicability but does not falsify its feasibility; what would falsify Diagnosis III is: a progressive multivariable intervention still undergoing a critical phase transition despite adequate implementation, or the relevant risks being merely transferred or deferred rather than reduced. Candidate cases:

As of 2026, no second case that can be clearly classified, within the comparative framework of this chapter, under the combination of "sustained governance + synchronized multivariable intervention + large-scale stock restructuring" has yet appeared in global financial history. The sustained structural treatment from 2017 to 2026 by China's New Asset Management Regulations of roughly RMB 100 trillion of shadow-banking stock is, among the cases currently included in this chapter's comparison, the one that most fully matches the model of Diagnosis III. This judgment is both the strength of Diagnosis III, since it demonstrates a hitherto unobserved institutional combination to be feasible, and its evidentiary weakness, because the sample size is n = 1. Diagnosis III is therefore formulated as an **open proposition**: if other economies produce cases over the next 10–20 years involving sustained governance + multivariable synchronization + large-scale stock restructuring, then structural

evidence for this diagnosis will either be strengthened or weakened accordingly.

The extension of Diagnosis III is that the relation between the governance persistence period and the post-manifestation settlement is re-understood as a governance dimension that can be designed rather than merely observed, and the AI-governance meaning of this extension will be developed in §4.

**Relation to the existing literature.** The core feature of Diagnosis III, the "multi-tool coordination + long transition period" of China's New Asset Management Regulations, has already been explicitly described in the existing Chinese literature on financial regulation. Yi Gang (2018), in his question-and-answer session at the Boao Forum for Asia, proposed "achieving a beautiful deleveraging" (author's translation) by keeping the debt level stable, optimizing the debt structure, and making the overall leverage ratio more reasonable. Guo Shuqing (2018), in his keynote speech at the Lujiazui Forum, stressed that forestalling and defusing financial risks must give full consideration to the capacity of institutions and markets, preventing the emergence of "the risk of handling risk" (author's translation), that is, avoiding a systemic crisis triggered by the intervention itself. Peng Wensheng (2017), in *Approaching the Financial Cycle*, characterized this policy orientation as "guiding the economy to digest in an orderly way the problems accumulated by excessive expansion of real estate and credit" (author's translation) and avoiding "disorderly, abrupt adjustment" (author's translation). Song & Xiong's (2018) analysis of the credit dual track and the expansion of shadow banking, and Chen, He & Liu's (2020) measurement of the see-saw between stimulus loans and shadow banking, provide background for understanding the policy environment of China's stock-compression period; "multi-tool coordination + long transition period" is this chapter's own characterization of the New Asset Management Regulations.

The relation of MCAF to these existing descriptions is one of **formalization and cross-case comparison**: "a beautiful deleveraging" (author's translation) corresponds in MCAF to "a design of the governance persistence period, measured in years rather than days"; "the risk of handling risk" (author's translation) corresponds to "controlling the slope of the manifestation function $M(\tau)$ so as to avoid a critical phase transition"; and "multi-tool coordination" corresponds to simultaneous intervention on the media conductivity coefficient $\Lambda$, number of nesting layers, narrative load $\phi$, implicit-guarantee narrative, and meta-audit $A$, look-through supervision.

The increment of MCAF lies in formalizing these qualitative descriptions into measurable variables, the duration of the governance persistence period, the slope of the manifestation function $M(\tau)$, and the directional change of media conductivity coefficient, narrative load, and meta-audit, and in comparing the Chinese case with the United States and Japan within a common coordinate system through the three-dimensional comparison table of §3.3.5. Existing literature is already quite sufficient in its policy narration of the New Asset Management Regulations; MCAF's contribution is to turn the notion of "progressive" from a qualitative style of policy into a **structural classification of governance-clock modes**, pulsed in the United States, sustained in China, and stagnant in Japan of the mid-1990s, and to show that this classification corresponds to different forms of the manifestation function $M(\tau)$ and different costs borne by the legitimacy base $\|L\|$.

---

## §4 Conclusion: MCAF's Meta-Contribution to Financial History and the Boundaries of the AI Analogy

The common structure of the three diagnoses in this chapter is that each extracts one structural observation about a particular MCAF variable combination from financial history and establishes its generality through

cross-case comparison. The clock ratios $\tau_g/\tau_c$ addressed by the three diagnoses all lie within the interval $10^0$-$10^1$, where the time-scale gap between regulation and market does not exceed one order of magnitude. Once the clock ratio is pushed to an extreme above $10^6$, as in high-frequency trading or AI generation speed, MCAF's intervention space contracts from ex post auditing to ex ante structural constraints—access restrictions, risk limits, and necessary rate limits (the financial counterpart of "precision speed-limiting" in Chapter 8 §7), with rate limitation being one tool among these rather than a single variable; and such constraints necessarily incur a cost in $\|L\|$. The AI significance of this structural constraint is discussed in Chapter 15, §4.

### §4.1 The meta-contribution of MCAF: an organizing principle rather than a new discovery

The increment of MCAF in this chapter **does not lie in discovering historical facts or policy mechanisms unknown to the existing literature**, but in **providing a structural language that makes cases comparable across cases**. This positioning is analogous to Mendeleev's periodic table: the elements had already been discovered before the periodic table, but the table provided a unified organizational principle that made their structural relations visible. The value of MCAF lies in using the six variables and three clocks to organize scattered cases from financial history into comparable structures, allowing "why defense support and stock disposal remained decoupled in the Japanese sample" and "why sustained governance could advance stock adjustment in the Chinese sample" to be understood as different variants of governance-clock mode within the same coordinate system—this chapter compares intervention timing and instrument combinations and makes no overall success-or-failure evaluation of the two countries' governance.

The incremental positioning of the three diagnoses is as follows:

- **Diagnosis I**, critical phase transition of the manifestation function: the core phenomenon, sudden explicitation of hidden risk, has already been precisely described in Gorton (2010) through "information sensitivity" and in Brunnermeier (2009) through the "liquidity spiral." The increment of MCAF is to formalize that phenomenon as a phase transition of $M(\tau)$ and to establish a causal chain between it and the accumulation window of $\tau_g/\tau_c \approx 10^{0.5}$-$10^1$, which allows Diagnosis I to be applied across time and geography.
- **Diagnosis II**, the impossibility of pure defense-side intervention: the core insight, that bank recapitalization without clearing bad assets is ineffective, was already clearly proposed in Caballero et al. (2008) through the zombie-lending model. The increment of MCAF is to elevate the policy-tool distinction between "capital injection vs. asset clearing" into a structural classification of "defense side, $A/\|L\|$, vs. projection side, $C$ inventory," and to establish the cross-case regularity that "touching $C$ inventory can shorten the post-manifestation settlement horizon."
- **Diagnosis III**, the progressive governance clock: the core feature, multi-tool coordination plus long transition period, has already been explicitly described by Yi Gang (2018), Guo Shuqing (2018), Peng Wensheng (2017), and others. The increment of MCAF is to formalize the "progressive" as a design of the governance persistence period and to compare the Chinese case with the United States and Japan within one common coordinate system through a three-dimensional comparison table.

**The real value of MCAF** lies in the fact that it does not discover new policy mechanisms, but **establishes a meta-language that makes different policy mechanisms structurally comparable**. Organizing cross-case comparison under the unified rubric of six variables and three clocks is the particular organization of this chapter—the existing literature already contains rich cross-country and cross-period research, this chapter

does not claim to have pioneered cross-case comparison, and its increment lies in the chart of accounts of the comparison itself.

### §4.2 Three fracture lines in the finance-AI analogy

The projection of the three diagnoses into AI is not unconditional. There are three structural fracture lines between finance and AI, and at specific points these fractures cause the analogy to fail:

**Fracture I: the absence of the compensation loop.** Section §2 of this chapter has shown the 258-year evolution of the financial compensation loop, from the Philadelphia mutual society in 1752 onward, as the financial system gradually built through crisis-driven institutional construction a three-layer structure of contractual compensation / systemic rescue / socialization of loss. AI has none of this structure at all, as Chapter 15 has already diagnosed. There is no mandatory cross-agent damage-compensation mechanism, no framework for systemic rescue, and no institutionalized loss-sharing arrangement—the existing sporadic vendor indemnity commitments and insurance products fall far short of an equivalent of the compensation loop in coverage, conditions of attribution, and funding assurance. The structural consequence of this absence is that finance's $M(\tau)$ can be compressed by $\tau_g$ after critical phase transition, as in the 18 days of TARP in 2008, precisely because the three-layer compensation loop already exists, the central bank as lender of last resort, FDIC deposit insurance, orderly liquidation in bankruptcy law, all institutional devices accumulated from 1913 to 2010. AI has none of these devices. This means that AI's first systemic crisis will have to be handled **without a compensation loop**. The slope of $M(\tau)$ may be steeper than in 2008, because there is no institutionalized buffer, and the compressive capacity of $\tau_g$ may be weaker, because there is no pre-stored emergency legislative procedure.

**Fracture II: the absence of an emergency governance mode.** In financial crisis there exists an emergency governance mode of compressed $\tau_g$, the United States passed TARP within 18 days in 2008, and crisis itself created the political window that allowed legislation normally measured in years to be completed on the scale of days. AI currently has no such mode. AI regulation is still measured in years, the EU AI Act took about three years from proposal to adoption, and there is still no comprehensive federal AI legislation in the United States. Even if AI generates a major harmful event, there is no precedent showing that it can trigger compression of $\tau_g$ comparable to a financial crisis. The compressive power of $\tau_g$ in finance depends upon the **immediate visibility** of financial collapse, bank failures, ATMs unable to dispense cash, stock-market crashes, all signals perceptible to the whole population within days or weeks. The visibility of AI harms may be far lower: algorithmic discrimination, filter bubbles, and degradation of cognitive capacity are harms that accumulate slowly over months or years, and do not possess the "visibility threshold" required to trigger emergency legislation.

**Fracture III: the difference in manifestation chains.** In the financial system, the manifestation of $D_c^{shadow}$ into $D_c^{obs}$ has **price signals as its manifestation chain**: falling equity prices, bond defaults, and spiking CDS spreads as price signals, together with bank runs and redemption waves as quantity-and-behavior signals, jointly constituting the multichannel signal structure through which financial risk manifests. Most AI harms lack market price as a manifestation chain, because they are externalities and are not reflected in the price system. Algorithmic discrimination will not produce some "discrimination index" crashing in the stock market; filter bubbles will not cause some "cognitive diversity index" to trigger margin calls. This means that AI's $M(\tau)$ may remain low for a very long time, with $D_c^{shadow}$ accumulating without manifestation, and then jump rapidly at some trigger point. Price signals both transmit risk information and amplify shocks through margin and fire-sale mechanics; AI harms lack such concentrated signals and may long remain difficult to attribute—but whether a

steeper concentrated manifestation occurs depends on propagation structure and trigger mechanisms, and cannot be derived from the absence of price signals alone. Before September 2008, finance already had multiple "warning signals", the rise in subprime default rates in 2007, the collapse of Bear Stearns hedge funds, the run on Northern Rock. These signals did not trigger systemic response, but they at least partially manifested at the price level. AI may lack such manifestation-chain warning signals. Its first systemic manifestation may directly take the form of a critical phase transition.

### §4.3 Three transplantable structures in the finance-AI analogy

Despite these three fracture lines, financial history still provides three transplantable structural insights for AI governance:

**Transplantable structure I: the typology of $\tau_g/\tau_c$.** The four clock scenarios established in §1 have direct counterparts in the AI scenario: AI's "normal accumulation window" corresponds to $\tau_g/\tau_c \approx 10^1$-$10^2$, model iteration measured in months while regulatory legislation is measured in years; AI's "extreme separation" corresponds to $\tau_g/\tau_c > 10^6$, model inference measured in milliseconds while regulatory response is measured in months, and this latter condition is even more common in AI. The transplantability of this typology lies in the fact that the order of magnitude of $\tau_g/\tau_c$ itself determines the available path of intervention. When $\tau_g/\tau_c > 10^6$, item-by-item manual auditing is physically incapable of catching the projection side, as already argued in §1.3; governance must turn to ex ante structural constraints and automated execution, with rate limitation being one tool among these rather than the only path.

**Transplantable structure II: the long-tail of the post-manifestation settlement.** Diagnosis II in §3.2 points out that compression of $\tau_g$, the U.S. passage of TARP in 18 days in 2008, does not shorten the post-manifestation settlement, which still required 10–15 years. This "clock non-synchronization" has a direct analogue in AI: even if AI regulation were to respond rapidly after some crisis, under a hypothetical emergency law, the long-tail payout of AI harms, degradation of cognitive ability, hardening of labor structure, erosion of social trust, would still continue on the scale of decades. The implication for AI governance is this: **do not confuse regulatory response speed with the duration of harm repair**. The former can be compressed in crisis; whether the latter is shortened depends on the timing, object, and manner of intervention.

**Transplantable structure III: the feasibility and cost of progressive governance.** Diagnosis III in §3.3 points out that when governance is designed as sustained and simultaneously intervenes on $\Lambda/\phi/A$, $M(\tau)$ can complete the manifestation of shadow debt without entering a critical phase-transition state. The AI analogue is that AI regulation can choose "progressive multidimensional intervention" rather than "waiting for crisis and then emergency response." The former requires that the governance persistence period be sustained on the scale of years or even a decade and that the regulatory architecture remain relatively independent of short-term industry pressure over the long term. The scope limitation is that China's New Asset Management Regulations are, among the cases currently included in this chapter's comparison, the one that most fully matches this pattern, n = 1, and their generalizability requires future testing in more cases. Even so, they provide AI governance with an **existence proof**: progressive governance is feasible in principle.

### §4.4 Scope and Limitations

The application of MCAF in the financial scenario cannot do the following things:

1. **It cannot predict the timing of crisis.** MCAF describes the structural form of $M(\tau)$, slope and critical conditions, not the specific time point of $M(\tau)$. It can say that "this structure makes critical phase transition possible"; it cannot say that "critical phase transition will occur on date X." Section §3.1.4 has already argued that the macro-structure of the phase transition of $M(\tau)$ is predictable, once the accumulation window of $\tau_g/\tau_c \approx 10^{0.5}$-$10^{1}$ continues long enough, phase transition will occur sooner or later, but its precise timing and trigger event are unpredictable. This is an inherent property of self-organized critical systems.
2. **It cannot precisely quantify $\Delta A$ and $\Delta\|L\|$.** The discussion of $\Delta A$ and $\Delta\|L\|$ in this chapter is qualitative and directional, up or down, not numerically precise. MCAF provides structural directional judgment rather than point estimation. This is the honest boundary of the framework.
3. **The finance-AI analogy is limited.** The three fracture lines above show that the experience of finance cannot be transplanted into AI unconditionally, especially along the three dimensions of the compensation loop, emergency governance mode, and manifestation chain. Any analogy from finance to AI must first test whether these three fractures invalidate the analogy.
4. **This chapter is a diagnostic tool, not a predictive tool.** The function of the three diagnoses is to "extract structural observations," not to "predict future events." Their value lies in establishing falsifiable structural propositions rather than providing operational policy advice. Moving from tool to recommendation requires additional institutional analysis and value judgment, and that is not the task of this chapter.

### §4.5 MCAF's meta-implication for AI

Chapter 15 has already used MCAF to conduct a dynamic calibration of the AI domain. Its three core diagnoses are: the absence of a manifestation chain for shadow debt (§15.3, the lack in AI of an institutionalized chain converting shadow debt into observable losses); structural lag of the defense side under the ordering of the three clocks (§15.4, the techno-capital cycle measured in months against a governance cycle measured in years, with AI governance not yet having formed the stable emergency-compression capability of the financial system's crisis mode); and institutional asymmetry at the individual level (§15.5, the organizational conditions for an independent oversight body being impossible to instantiate within an individual cognitive system). This chapter serves as the **institutional comparison baseline** for Chapter 15. The 258 years of institutional evolution in financial history provide an order-of-magnitude reference for AI's condition of "not yet having begun accumulation." Together the two chapters form one set of comparative tests of the MCAF framework: in a domain that has already evolved a complete compensation loop, finance, what new can MCAF say? In a domain where the compensation loop has not yet even begun to evolve, AI, how severe is the diagnosis MCAF makes?

The final conclusion of this chapter is not that "AI should learn from financial regulation." The three fracture lines invalidate that simple analogy. The final conclusion is that **financial history provides a complete sample of how cognitive debt accumulates, manifests, and is liquidated on the scale of centuries**, and this sample allows AI governance to orient itself by reference to a known structural trajectory rather than groping in total darkness. The value of MCAF does not lie in telling AI governors what they "should do," but in providing a **meta-language that makes the structural consequences of different choices comparable**. That meta-language makes it possible to compare "progressive governance vs. pulsed governance," "defense-side intervention vs. projection-side intervention," and "$M(\tau)$ phase transition vs. linear manifestation" between financial history and AI's future. That is this chapter's contribution to the book as a whole: not financial history itself, but **a structural language that allows financial history to enter into dialogue with AI's future**.

## References


21st Century Business Herald. (2019). The Defaults of the Year: Bond Extensions, Interbank Defaults, and What Else Is Possible? December 31. (Li Yumin, Xue Ruyun.) (In Chinese.) https://m.21jingji.com/article/20191231/herald/e07bab7dc9e0982d8ad5c521a822e3c7.html

Bak, P. (1996). *How Nature Works: The Science of Self-Organized Criticality*. Copernicus.

Benmelech, E. & Dlugosz, J. (2009). The Alchemy of CDO Credit Ratings. *Journal of Monetary Economics*, 56(5). (NBER Working Paper No. 14878; Table 2 "Global CDO Issuance Volume" reproduces SIFMA statistics.) https://doi.org/10.1016/j.jmoneco.2009.04.007

Born, B. (1998). Testimony Concerning the Over-the-Counter Derivatives Market before the U.S. House Committee on Banking and Financial Services. July 24, 1998. https://www.cftc.gov/sites/default/files/opa/speeches/opaborn-33.htm

Brevoort, K. P. & Cooper, C. R. (2013). Foreclosure's Wake: The Credit Experiences of Individuals Following Foreclosure. *Real Estate Economics*, 41(4). https://doi.org/10.1111/1540-6229.12021

Brunnermeier, M. K. (2009). Deciphering the Liquidity and Credit Crunch 2007–2008. *Journal of Economic Perspectives*, 23(1). https://doi.org/10.1257/jep.23.1.77

Caballero, R. J., Hoshi, T. & Kashyap, A. K. (2008). Zombie Lending and Depressed Restructuring in Japan. *American Economic Review*, 98(5). https://doi.org/10.1257/aer.98.5.1943

Caixin. (2022). Nationwide Special-Purpose Loan of RMB 200 Billion for "Ensuring Housing Delivery" Launched. August 29. (In Chinese.) https://other.caixin.com/2022-08-29/101932575.html

Chen, Z., He, Z. & Liu, C. (2020). The Financing of Local Government in China: Stimulus Loan Wanes and Shadow Banking Waxes. *Journal of Financial Economics*, 137(1). https://doi.org/10.1016/j.jfineco.2019.07.009

China Banking Association. (2023). *China Banking Wealth-Management Business Development Report and Wealth-Management Company Yearbook (2023)*. Beijing. (In Chinese.) http://www.chinabanking.com.cn/zixun/zixun_xq?9503

China Wealth Management Registration and Depository Center. (2021). *China Banking Wealth-Management Market Semi-Annual Report (2021 H1)*. Beijing. (In Chinese.) https://www.chinawealth.com.cn

China Wealth Management Registration and Depository Center. (2022). *China Banking Wealth-Management Market Annual Report (2021)*. Beijing. (In Chinese.) https://www.chinawealth.com.cn

China Wealth Management Registration and Depository Center. (2023). *China Banking Wealth-Management Market Annual Report (2022)*. Beijing. (In Chinese.) https://www.fundmarket.com.cn/upfiles/attachments//638143183983655067169 1993.pdf

China Wealth Management Registration and Depository Center. (2024). *China Banking Wealth-Management Market Annual Report (2023)*. Beijing. (In Chinese.) https://www.chinawealth.com.cn

China Wealth Management Registration and Depository Center. (2026). *China Banking Wealth-Management Market Annual Report (2025)*. Beijing. (In Chinese.) https://www.chinawealth.com.cn

Chopra, A., Kang, K., Karasulu, M., Liang, H., Ma, H. & Richards, A. (2002). From Crisis to Recovery in Korea: Strategy, Achievements, and Lessons. In D. T. Coe & S.-J. Kim (Eds.), *Korean Crisis and Recovery*. IMF and Korea Institute for International Economic Policy. https://www.elibrary.imf.org/downloadpdf/display/book/9781589060685/C02.pdf

Commodity Futures Trading Commission (CFTC). (1998-05-07). *CFTC Issues Concept Release Concerning Over-the-Counter Derivatives Market*. Press Release 4142–98. https://www.cftc.gov/sites/default/files/opa/press98/opa4142-98.htm

Cordell, L., Huang, Y. & Williams, M. (2012). Collateral Damage: Sizing and Assessing the Subprime CDO Crisis. Federal Reserve Bank of Philadelphia Working Paper No. 11–30/R. https://www.philadelphiafed.org/-/media/frbp/assets/working-papers/2011/wp11-30r.pdf

Dinç, I. S. & McGuire, P. M. (2004). Did Investors Regard Real Estate as "Safe" during the "Japanese Bubble" in the 1980s? *BIS Working Papers* No. 164. Bank for International Settlements. https://www.bis.org/publ/work164.htm

Edwards, F. R. (1999). Hedge Funds and the Collapse of Long-Term Capital Management. *Journal of Economic Perspectives*, 13(2). https://doi.org/10.1257/jep.13.2.189

*Encyclopedia of Chicago*. "Insurance." Chicago Historical Society / Newberry Library. https://encyclopedia.chicagohistory.org/pages/645.html

Englund, P. (1999). The Swedish Banking Crisis: Roots and Consequences. *Oxford Review of Economic Policy*, 15(3). https://academic.oup.com/oxrep/article-abstract/15/3/80/381439

FDIC (Federal Deposit Insurance Corporation). (1984). *The First Fifty Years: A History of the FDIC, 1933–1983*, Chapter 3: Establishment of the FDIC. https://www.fdic.gov/resources/publications/first-fifty-years/book/first-fifty-chapter3.pdf

FDIC (Federal Deposit Insurance Corporation). (1998). *A Brief History of Deposit Insurance in the United States*. FDIC. https://www.fdic.gov/bank/historical/brief/brhist.pdf

Financial Services Agency (FSA). (2009). Transition of Non-Performing Loans Based on the Financial Reconstruction Act. https://www.fsa.go.jp/en/regulated/npl/20090807/01.xlsx

Financial Services Commission of Korea. (2009). Public-Fund Injection and Recovery Status (as of end-June 2009). Xinhua News Agency dispatch, Seoul, August 17. https://english.cctv.cn/20090818/100977.shtml

Fradkin, P. L. (2005). *The Great Earthquake and Firestorms of 1906: How San Francisco Nearly Destroyed Itself*. University of California Press.

Gorton, G. B. (2010). *Slapped by the Invisible Hand: The Panic of 2007*. Oxford University Press.

Greenspan, A. (2008). Testimony before the U.S. House Committee on Oversight and Government Reform. *The Financial Crisis and the Role of Federal Regulators*, Serial No. 110–209, October 23, 2008. https://www.govinfo.gov/content/pkg/CHRG-110hhrg55764/html/CHRG-110hhrg55764.htm

Guo Shuqing. (2018). Preventing Risk Requires "Treating Illness Before It Arises." *People's Daily Online – People's Daily*, June 15. (In Chinese.) https://finance.people.com.cn/n1/2018/0615/c1004-30060249.html

Haggerty, M. (2022). Lessons Learned: Brooksley Born. *Journal of Financial Crises*, 4(2). https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss2/91

Hayashi, F. & Prescott, E. C. (2002). The 1990s in Japan: A Lost Decade. *Review of Economic Dynamics*, 5(1). https://doi.org/10.1006/redy.2001.0149

Hipple, S. F. (2016). Labor Force Participation: What Has Happened since the Peak? *Monthly Labor Review*, U.S. Bureau of Labor Statistics. https://doi.org/10.21916/mlr.2016.43

Hoshi, T. & Kashyap, A. K. (2010). Will the U.S. Bank Recapitalization Succeed? Eight Lessons from Japan. *Journal of Financial Economics*, 97(3). https://doi.org/10.1016/j.jfineco.2010.02.005

International Monetary Fund. (2025). *World Economic Outlook Database*. https://www.imf.org/external/ns/cs.aspx?id=28

International Swaps and Derivatives Association (ISDA). (2010). *ISDA Market Survey: Mid-Year 2010*. https://www.isda.org/a/j3DDE/isda-market-surveys-2010-1995.docx

Investment Company Institute (ICI). (2009). *Report of the Money Market Working Group*. Washington, DC: ICI. https://www.ici.org/pubfile_pdf/ppr_09_mmwg.pdf

Ito, T. & Mishkin, F. S. (2004). Two Decades of Japanese Monetary Policy and the Deflation Problem. NBER East Asia Seminar on Economics (EASE15). https://www.nber.org/papers/w10878

Kacperczyk, M. & Schnabl, P. (2013). How Safe Are Money Market Funds? *Quarterly Journal of Economics*, 128(3). https://doi.org/10.1093/qje/qjt010

Koo, R. C. (2008). *The Holy Grail of Macroeconomics: Lessons from Japan's Great Recession*. John Wiley & Sons.

Lowenstein, R. (2000). *When Genius Failed: The Rise and Fall of Long-Term Capital Management*. Random House.

MacKenzie, D. (2006). *An Engine, Not a Camera: How Financial Models Shape Markets*. MIT Press.

MarketWatch. (2008). Lehman CDS spreads wider than Bear's before bailout: CDR. September 11. https://www.marketwatch.com/story/lehman-cds-spreads-wider-than-bears-before-bailout-cdr

McDonald, R. L. & Paulson, A. (2014). AIG in Hindsight. Federal Reserve Bank of Chicago Working Paper No. 2014-07. https://www.chicagofed.org/~/media/publications/working-papers/2014/wp2014-07-pdf.pdf

Microsoft. (2023). Microsoft Announces New Copilot Copyright Commitment for Customers. *Microsoft On the Issues*, September 7. https://blogs.microsoft.com/on-the-issues/2023/09/07/copilot-copyright-commitment-ai-legal-concerns/

Miller, N. C. (1965). *The Great Salad Oil Swindle*. New York: Coward-McCann.

Moody's Investors Service. (2018). *China Shadow Banking Monitor—Quarterly Report, December 2018*. (Archived in the local literature-replication folder for this chapter.)

National Bureau of Statistics. (various years). *Statistical Communiques on National Economic and Social Development*. Beijing. (In Chinese.) https://www.stats.gov.cn/sj/tjgb/ndtjgb/

National Financial Regulatory Administration. (2025). *Measures for the Administration of Information Disclosure of Asset-Management Products of Banking and Insurance Institutions* (Order No. 10 of 2025). (In Chinese.) https://www.nfra.gov.cn/cn/view/pages/rulesDetail.html?docId=1239473

Odell, K. A. & Weidenmier, M. D. (2004). Real Shock, Monetary Aftershock: The 1906 San Francisco Earthquake and the Panic of 1907. *Journal of Economic History*, 64(4). https://www.jstor.org/stable/3874987

Payan, J. (2013). A New Look at the Great Chicago Fire. *Rough Notes*, October 2013. https://roughnotes.com/rnmagazine/2013/october/2013_10p048.htm

Peek, J. & Rosengren, E. S. (2005). Unnatural Selection: Perverse Incentives and the Misallocation of Credit in Japan. *American Economic Review*, 95(4). https://doi.org/10.1257/0002828054825691

Peng Wensheng. (2017). *Approaching the Financial Cycle: Money, Credit, and the Distance between the Real and the Virtual*. CITIC Press. (In Chinese.)

People's Bank of China, China Banking and Insurance Regulatory Commission, China Securities Regulatory Commission, & State Administration of Foreign Exchange. (2018). *Guiding Opinions on Regulating the Asset Management Business of Financial Institutions* (Yin Fa [2018] No. 106). (In Chinese.) https://www.gov.cn/zhengce/zhengceku/2018-12/31/content_5433072.htm

People's Bank of China Financial Stability Analysis Group. (2019). *China Financial Stability Report 2019*. Beijing: China Financial Publishing House. (In Chinese.) http://www.pbc.gov.cn/goutongjiaoliu/113456/113469/3927456/2019112607514924977.pdf

People's Daily Online. (2022). Multiple Departments Introduce Measures to Advance "Ensuring Housing Delivery, Stabilizing Livelihoods" Work. August 23. (In Chinese.) http://finance.people.com.cn/n1/2022/0823/c1004-32508661.html

Pozsar, Z., Adrian, T., Ashcraft, A. & Boesky, H. (2013). Shadow Banking. Federal Reserve Bank of New York Staff Report No. 458. https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr458.pdf

Rajan, R. G. (2005). Has Financial Development Made the World Riskier? *Proceedings – Economic Policy Symposium – Jackson Hole*, Federal Reserve Bank of Kansas City. https://www.kansascityfed.org/Jackson%20Hole/documents/3326/PDF-Rajan2005.pdf

Research Group of the Policy Research Bureau of the CBIRC. (2020). China Shadow Banking Report. *Financial Regulation Research*, 2020(11). (In Chinese.)

Resolution Trust Corporation. (1995). *Annual Report of the Resolution Trust Corporation 1995*. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/ar_rtc_1995.pdf

Schmidt, L., Timmermann, A. & Wermers, R. (2016). Runs on Money Market Mutual Funds. *American Economic Review*, 106(9). https://doi.org/10.1257/aer.20140678

Schularick, M. & Taylor, A. M. (2012). Credit Booms Gone Bust: Monetary Policy, Leverage Cycles and Financial Crises, 1870–2008. *American Economic Review*, 102(2). https://doi.org/10.1257/aer.102.2.1029

Seligman, J. (1982). *The Transformation of Wall Street: A History of the Securities and Exchange Commission and Modern Corporate Finance*. Houghton Mifflin.

Song, Z. M. & Xiong, W. (2018). Risks in China's Financial System. *Annual Review of Financial Economics*, 10. https://doi.org/10.1146/annurev-financial-110716-032402

Sorkin, A. R. (2009). *Too Big to Fail: The Inside Story of How Wall Street and Washington Fought to Save the Financial System—and Themselves*. Viking.

Statistics Bureau of Japan. (2012). *2012 Employment Status Survey*. (In Japanese.) https://www.stat.go.jp/data/shugyou/2012/index.html

Sufi, A. & Taylor, A. M. (2022). Financial Crises: A Survey. In G. Gopinath, E. Helpman & K. Rogoff (Eds.), *Handbook of International Economics*, Vol. 6. Elsevier. https://www.sciencedirect.com/science/chapter/handbook/pii/S1574440422000120

Valukas, A. R. (2010). *Lehman Brothers Holdings Inc. Chapter 11 Proceedings Examiner's Report*. United States Bankruptcy Court, Southern District of New York. https://www.jenner.com/en/lehman-brothers-holdings-inc-chapter-11-proceedings-examiner-report

Xiao Gang. (2020). Advancing the Transformation of China's Asset-Management Industry in an Orderly Way. Keynote speech at the 2020 China Asset Management Annual Conference, August 9. (In Chinese.) https://www.cs.com.cn/xwzx/hg/202008/t20200809_6083821.html

Yi Gang. (2018). China Will Achieve Deleveraging through Three Major Tasks. *Sina Finance (Xinhua News Agency dispatch)*, April 11. (In Chinese.) https://finance.sina.com.cn/roll/2018-04-11/doc-ifyteqtq8257274.shtml

Zhang Yilin, Liang Wei & Yu Yunjun. (2023). The Formation and Governance Mechanism of Rigid Payment in the Development of China's Shadow Banking. *Economic Research Journal*, 2023(2). (In Chinese.) http://erj.ajcass.com/

Zheng, J. & Meister, M. (2025). The Unbearable Slowness of Being: Why Do We Live at 10 Bits/s? *Neuron*, 113(2). https://doi.org/10.1016/j.neuron.2024.11.008

Zhong Zhen. (2018). Financial Regulatory Coordination: The Proposal and Application of Channel Theory. *Finance & Trade Economics*, 39(9). (In Chinese.) https://cmjj.ajcass.com/Magazine/Show?ID=88962